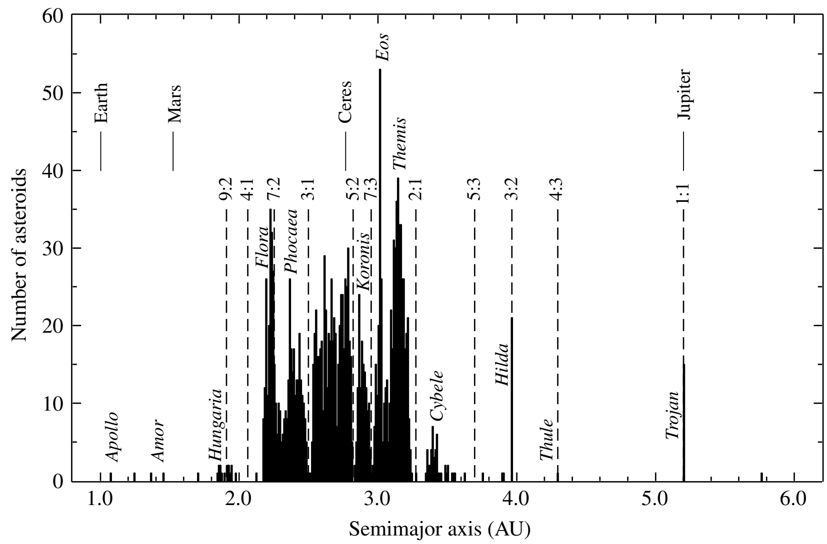


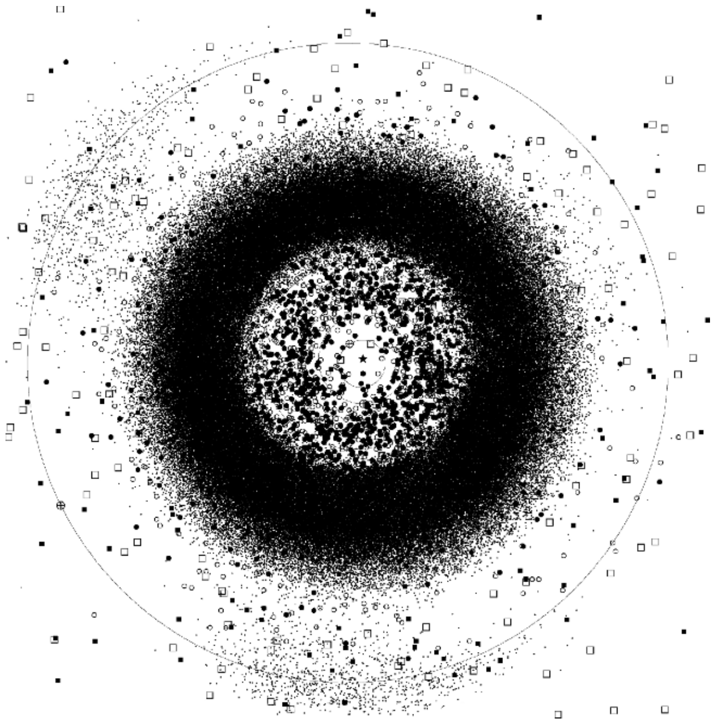
FIGURE 22.13 The distribution of 1796 asteroids in the asteroid belt. Asteroid group names and orbital resonances with Jupiter are also shown. Kirkwood gaps are evident at numerous resonance locations, and enhancements in the number of asteroids are apparent at other resonance locations. (Data from Williams, *Asteroids II*, Binzel, Gehrels, and Matthews (eds.), University of Arizona Press, Tucson, 1989.)

The Kirkwood Gaps in the Asteroid Belt

The distribution of asteroids in the belt is not completely uniform or even smoothly varying with distance from the Sun. Instead, for various values of the orbital semimajor axis, asteroids are either conspicuously absent or overabundant (see Fig. 22.13). These positions correspond to orbital resonances with Jupiter, analogous to the resonances in Saturn's rings that are produced by its moons, most notably Mimas. Regions where asteroids are underabundant are known as the **Kirkwood gaps**, the most prominent being at 3.3 AU (a 2:1 resonance of orbital periods) and at 2.5 AU (a 3:1 resonance). In reality, physical gaps in the belt, equivalent to gaps in Saturn's rings such as the Cassini division, do not actually exist. Instead, the varying eccentricities and orbital inclinations of the asteroids tend to smear out the gaps somewhat, populating them with objects that are transients at those radii. The locations of the asteroids (and of some comets) on September 8, 2005, projected onto the plane of the ecliptic, are shown in Fig. 22.14.

The Trojan Asteroids

In some cases, resonances with Jupiter correspond to local increases in the number of asteroids. A particularly interesting resonance group is the **Trojan asteroids** (1:1), which occupy the same orbit as Jupiter but either lead or trail the planet by 60°, as illustrated in Fig. 22.15 and evident in Figs. 22.11 and 22.14. In addition, at least one asteroid is orbiting the Sun at the trailing 60° position in Mars's orbit, and two are in the lead 60° position



Plot prepared by the Minor Planet Center (2005 Sept 8).

FIGURE 22.14 The distribution of minor bodies in the inner Solar System on September 8, 2005, projected onto the ecliptic. Approximately 237,000 objects are shown in this plot. The outer orbit is that of Jupiter, and the asteroid belt is clearly visible. The belt is not actually saturated with asteroids; rather, the symbols representing them are vastly larger than the objects themselves. The location of each planet is marked by a $\oplus$ sign; Jupiter is seen in the lower left of the diagram. Jupiter's Trojan asteroids are evident in the large "clouds" that lead and trail Jupiter by 60°. The orbits of the terrestrial planets are visible among the clutter of the Amors, Apollos, and Atens. Comets are indicated by squares, as in Fig. 22.11. In this orientation the planets and most of the other objects orbit counterclockwise, and the vernal equinox is to the right. (Courtesy of Gareth Williams, Minor Planet Center.)

of Neptune's orbit. These asteroids are found in regions of unusual gravitational stability (gravitational "wells") that are established by the combined influence of the Sun and Jupiter. The positions are the L_4 and L_5 Lagrangian points, which are locations of equilibrium that exist in a three-body system when one of the bodies (in this case an asteroid) is much smaller than the other two. Recall that Lagrangian points were discussed in detail in Section 18.1; they play an important role in the evolution of some binary star systems.

The Amors, Apollos, and Atens

Other special groups of asteroids are those that have orbits among the terrestrial planets. The **Amors** are located between the orbits of Mars and Earth, the **Apollos** cross Earth's orbit as they approach perihelion, and the **Atens** have semimajor axes that are less than 1 AU,

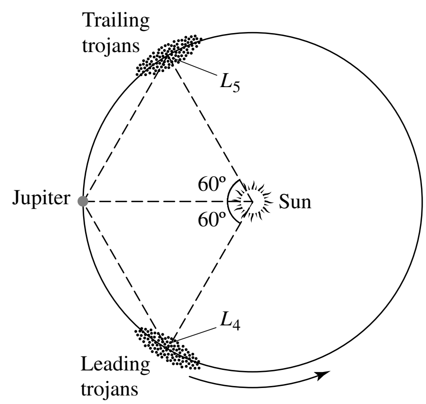


FIGURE 22.15 Trojan asteroids are located in Jupiter's orbit, either leading or trailing the planet by 60°. The occupied positions are two of the five Lagrangian points in the Sun–Jupiter system.

although they can cross Earth's orbit near aphelion. It appears that many of these objects were probably main-belt asteroids at one time, but perturbations with Jupiter reoriented their orbits. Some of the Earth-crossing objects could also be extinct cometary nuclei that have lost most of their volatiles after repeated trips near the Sun. Since the Apollo and Aten asteroids intersect Earth's orbit, there is always the possibility that a collision could occur.

Hirayama Families

In 1918, the Japanese astronomer Kiyotsugu Hirayama (1874–1943) pointed out associations of asteroids that occupy nearly identical orbits. Today, more than 100 **Hirayama families** (also known as **asteroid families**) have been identified. It is believed that each family was once a single larger asteroid that suffered a catastrophic collision. With collision speeds that can reach 5 km s^{-1}, the available energy is more than enough to crush rock and cause pieces of the original asteroid to escape. If the collisional energy is not sufficient, only a portion of the surface may escape, or, after fracture, the self-gravity of the debris could cause the asteroid to reform again as a rubble pile. The Infrared Astronomical Satellite observed dust bands that seem to be associated with some of the major Hirayama families.

Rendezvousing with Asteroids

The first asteroids to be visited by a spacecraft were 951 Gaspra and 243 Ida in 1991 and 1993, respectively (Ida and its moon, Dactyl, are shown in Fig. 22.12). The flybys occurred as the Galileo spacecraft passed through the asteroid belt while on its trip to Jupiter. Just as astronomers had expected, the irregularly shaped asteroids show evidence of having sustained numerous meteoritic impacts throughout their existence. In fact, the number of impacts suggests that Gaspra (a member of the Flora family) was probably broken off from a larger asteroid 200 million years ago. On the other hand, age estimates for Ida (a member of the Koronis family) vary. Given the small size of Dactyl, it is unlikely that the moon could have existed for more than 100 million years without getting destroyed by a major collision. However, based on the high crater density on Ida's surface, it appears that Ida may be as old as 1 billion years. Assuming that the two objects were created together from the breakup of a larger body, the resolution to the puzzle may rest with an increased rate of cratering from debris created when the larger object was destroyed. Even though the number density of asteroids in the belt is very low, the expected frequency of collisions is such that very few would have been lucky enough to avoid a major impact sometime during the Solar System's history.

With the discovery of Dactyl orbiting Ida, it is possible to estimate the mass of Ida from Kepler's third law. Unfortunately, because of the high relative speed of the flyby ($12.4\,\mathrm{km\ s^{-1}}$) and the spacecraft's trajectory relative to the orbit of Dactyl (the angle between the trajectory and the orbit was about 8°), only an approximate range of orbits were derived from the data. The results suggest that the mass of Ida is approximately 3 to 4×10^{16} kg, giving an average density of between 2200 and 2900 kg m^{-3}.

The **Near Earth Asteroid Rendezvous mission** (NEAR–Shoemaker)[13] was launched in 1996. On its way to its ultimate destination of asteroid 433 Eros, NEAR–Shoemaker also made a 10-km s^{-1} flyby of 253 Mathilde. Based on the gravitational perturbations of Mathilde on the spacecraft, it was determined that the average density of the asteroid is only 1300 kg m^{-3}, indicating that this asteroid is likely a very heavily fractured rubble pile that has been broken apart by multiple collisions and only loosely reassembled by its own gravity.

When NEAR–Shoemaker arrived at 433 Eros on February 14, 2000, it entered into orbit around the asteroid and began a year-long intensive study of the object (see Fig. 22.16).

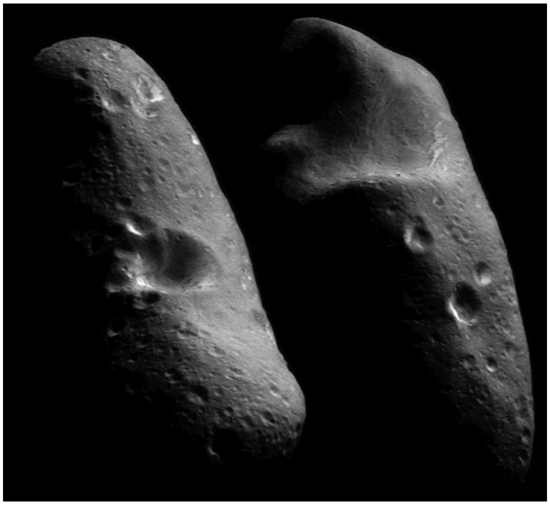


FIGURE 22.16 A composite image of the two hemispheres of 433 Eros as observed from orbit around the asteroid. Eros is heavily covered with regolith and shows significant evidence of cratering. Eros is one of the largest near-Earth asteroids, measuring 33 km long by 8 km wide by 8 km thick. (Courtesy of NASA/Johns Hopkins University Applied Physics Laboratory.)

[13]The mission was renamed in flight in honor of the late Eugene M. Shoemaker, planetary scientist and co-discoverer of the Shoemaker–Levy 9 comet. Shoemaker had always said that he wanted to hit 433 Eros with a rock hammer to see what was inside.

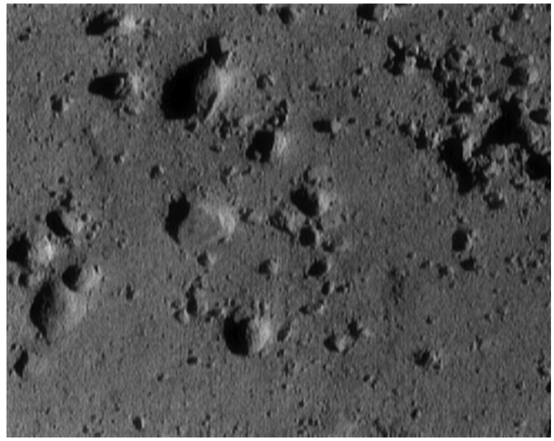


FIGURE 22.17 The surface of Eros from an altitude of 250 m. The image is 12 m across. The image was taken during the February 12, 2001, descent of the NEAR–Shoemaker spacecraft. (Courtesy of NASA/Johns Hopkins University Applied Physics Laboratory.)

During the orbital mission, the spacecraft studied Eros's gravitational field and obtained information about its surface composition. Although not designed as a lander, after one year in orbit, NEAR–Shoemaker survived an intentional landing on the surface of the asteroid at a speed of about 1.6 m s^{-1}, and was able to transmit information back to Earth for another week. During the descent phase of the mission, NEAR–Shoemaker returned many close-up images, including the one shown in Fig. 22.17.

Measurements obtained during the mission indicate that the density of Eros is 2670 kg m^{-3} and that it has probably been fractured, but not to the point of being a rubble pile like Mathilde. The interior of Eros appears to have a porosity of 25% or so. From measurements of radioactivity and from gamma-ray spectroscopy measurements (see Fig. 22.18), it appears that Eros contains K, Th, U, Fe, O, Si, and Mg, as expected for this primitive object.

Classes of Asteroids

It was in the 1930s that astronomers first realized that asteroids vary in color. By observing the spectrum of reflected sunlight, it is possible to identify absorption bands that provide important information about the surface compositions of these objects. Information can also be obtained by studying their albedos. The composition of asteroids is now known to vary significantly, but a general trend exists with increasing distance from the Sun (see Fig. 22.19). Some of the major classes of asteroids are

- **S-type** asteroids reside in the inner part of the belt (2–3.5 AU) and make up roughly one-sixth of all the known asteroids. Their surfaces are dominated by a mixture of iron- or magnesium-rich silicates, together with pure metallic iron–nickel. They tend to have a low abundance of volatiles, appear somewhat reddish, and have moderate albedos (0.1–0.2). Gaspra, Ida, and Eros are S-type asteroids.

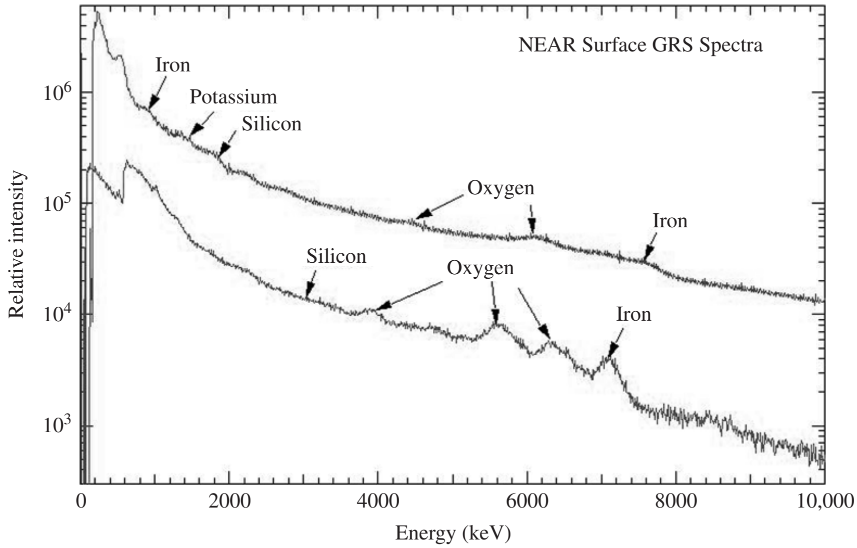


FIGURE 22.18 The gamma-ray spectrum of 433 Eros obtained by the NEAR–Shoemaker spacecraft after it landed on the surface of the asteroid. Two different gamma-ray detectors obtained the two spectra shown. (Adopted from a figure courtesy of NASA/Johns Hopkins University Applied Physics Laboratory.)

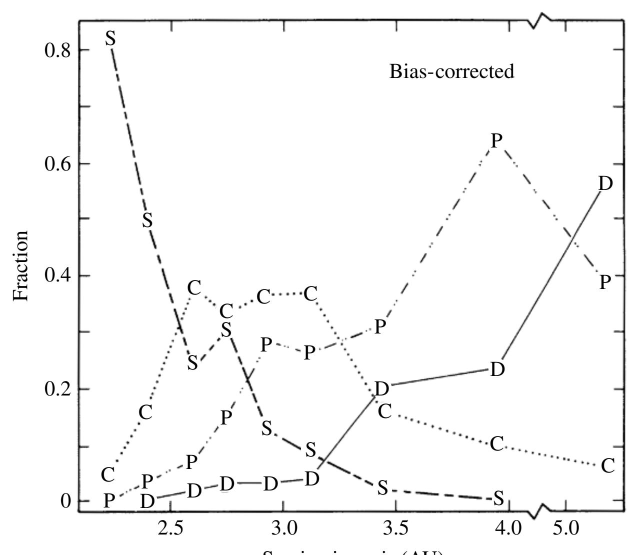


FIGURE 22.19 The distribution of major asteroid types with distance from the Sun. [Figure adapted from Gradie, Chapman, and Tedesco, *Asteroids II*, Binzel, Gehrels, and Matthews (eds.), University of Arizona Press, Tucson, 1989.]

- **M-type** asteroids are very metal-rich with absorption spectra dominated by iron and nickel. They appear slightly reddish, and they have moderate albedos (0.10–0.18). M-types are preferentially located in the inner portion of the belt, among the S-types (2–3.5 AU).

- **C-type** asteroids constitute perhaps three-fourths of all the minor planets. These objects are located predominantly near 3 AU but can be found throughout the main belt (2–4 AU). They are very dark, with albedos in the range 0.03–0.07, and they appear to be rich in carbonaceous material. Two-thirds of the C-types also contain significant quantities of volatiles, particularly water. Mathilde is a C-type asteroid.

- **P-type** asteroids are located near the outer edge of the main belt and beyond (3–5 AU), peaking in population near 4 AU. They have a slightly reddish appearance and low albedos (0.02–0.06). Their surfaces may contain a significant abundance of ancient organic compounds, which are also present in comets.

- **D-type** asteroids are much like P-types, except that they have a redder appearance and are located farther from the Sun. The Trojan asteroids are dominated by D-types. Some of Jupiter's smaller moons also exhibit spectra similar to D-type asteroids.

It seems likely that the differences in asteroid types with distance from the Sun are largely a result of the process of condensation out of the solar nebula (to be discussed in more detail in Section 23.2). Closer to the Sun, near the inner edge of the asteroid belt where the temperature was higher, more refractory compounds (like silicon) condensed out while volatiles such as water and organic compounds could not. Farther from the Sun, temperatures had decreased sufficiently to allow more volatile compounds to condense and become part of the asteroids in that region. Many of the C-type asteroids in the middle of the belt appear to be *hydrated* (meaning that water is present in these objects), whereas the more distant P- and D-types may contain water-ice, like most of the moons of the outer Solar System. Interestingly, 1 Ceres, at a distance from the Sun of 2.77 AU, and the largest asteroid in the belt, appears to be nearly spherical and may have a water-ice mantle.

Evidently the majority of the minor planets in the inner part of the belt have been subjected to significant gravitational separation during their lifetimes, including most or all of the S-type asteroids. The unusual and very metal-rich M-types have also been profoundly altered by evolution since formation. It is generally believed that M's represent the cores of much larger parent asteroids that became chemically differentiated and were later shattered by cataclysmic collisions, exposing the core.

At least one asteroid, 4 Vesta, appears to have a surface that is covered with basalt (rock formed from lava flows). Vesta has a radius of 250 km and is the third-largest asteroid known, behind 1 Ceres and 2 Pallas. It seems that magma developed in the interior and eventually found its way through cracks to the surface, where it solidified. Vesta also has an impact crater large enough to have exposed the subsurface mantle.

Internal Heating

As Vesta and the S- and M-type asteroids suggest, the interiors of at least some asteroids must have become molten for a period of time during their lives, raising the question of the source of the heat. Being small objects, the asteroids readily radiate their interior heat into space, so they should have cooled off rather quickly after formation, too quickly to allow for significant gravitational separation (recall that $\tau_{\rm cool} \propto R$; see Section 21.1). Furthermore, the very long-half-life radioactive isotopes that are, in large part, responsible for maintaining the hot interior of Earth could not generate heat rapidly enough to melt the interior of an asteroid. It has been suggested that a relatively short, intense burst of heat could be produced if a shorter-half-life isotope were available in sufficient abundance. A likely candidate is $^{26}_{13}\mathrm{Al}$, with a half-life of 716,000 years:

$$^{26}_{13}\mathrm{Al} \rightarrow {}^{26}_{12}\mathrm{Mg} + e^+ + \nu_e. \tag{22.5}$$

One difficulty with this suggestion is that in order to be effective in melting the interior of an asteroid, the aluminum must be incorporated relatively rapidly into a forming asteroid after the aluminum is produced (in just a small number of half-lives). This places a severe constraint on the formation timescale for the Solar System.

A second problem with the radioactive isotope solution lies in the apparent trend from chemically differentiated, volatile-poor asteroids in the inner belt to hydrated asteroids around 3.2 AU and icy bodies near Jupiter. This distribution seems to imply that $^{26}_{13}\mathrm{Al}$ was preferentially included in asteroids in the inner belt if it is the source of heat that led to chemical differentiation.

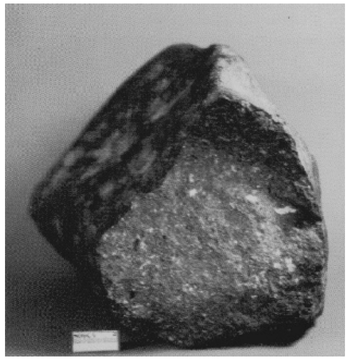
22.4 ■ METEORITES

In the early morning hours of February 8, 1969, residents in the region around Chihuahua City, Mexico, saw a bright blue-white light that streaked across the sky. As they watched, the light broke into two parts, each in turn exploding into a spectacular display of glowing fragments. Sonic booms were also heard accompanying the light show. It was reported that some observers even believed that the world was coming to an end. Rocks rained down on the countryside over an area that measured 50 km by 10 km (known as a **strewnfield**). The next day the first meteorite was discovered in the small village of Pueblito de Allende. All of the more than two tons of specimens collected from this meteor shower are now collectively referred to as the **Allende meteorite**. Many of the Allende stones were taken to the NASA Lunar Receiving Laboratory in Houston, Texas, for study.[14] One sample of the Allende meteorite is shown in Fig. 22.20.[15]

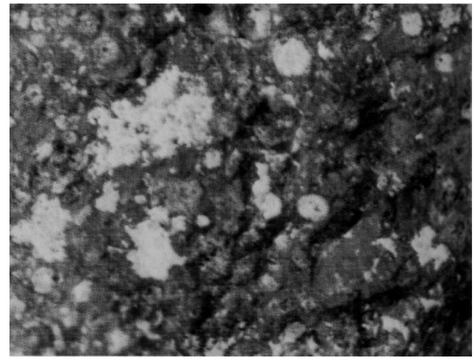
The observed streaks of light were produced by the frictional heating of the meteorite surfaces by Earth's atmosphere, causing the meteorites to glow. Although the outsides of the samples were covered by **fusion crusts** produced by the frictional heating, the interiors of the samples were unaffected. When a meteorite passes through the atmosphere, its damaged surface flakes off almost as quickly as it forms.

[14]The Lunar Receiving Laboratory was preparing to analyze the Moon rocks that were to be collected later that year by the *Apollo* astronauts.

[15]After attending a lecture about a meteoritic fall given by two Yale professors in Connecticut in 1807, President Thomas Jefferson (1743–1826) reportedly commented, "I could more easily believe that two Yankee professors could lie than that stones could fall from Heaven." Jefferson was, himself, a well-respected amateur scientist.



(a)



(b)

FIGURE 22.20 (a) A sample of the Allende meteorite. The surface has a fusion crust. (b) A close-up of a portion of the interior of the sample showing CAIs and chondrules embedded in a matrix. (Courtesy of Smithsonian Astrophysical Observatory.)

The Age and Composition of the Allende Meteorite

A very precise chronometer for determining ages of events in the formation of the Solar System is available by comparing the relative abundances of two stable isotopes of lead that can be identified in meteorites, $^{207}_{82}$Pb and $^{206}_{82}$Pb. These isotopes are ultimately produced by independent sequences of decays that begin with $^{235}_{92}$U (half-life of 0.704 Gyr) and $^{238}_{92}$U (half-life of 4.47 Gyr), respectively. By using this **Pb–Pb system**, scientists have deduced an age for the Allende meteorite of 4.566 ± 0.002 Gyr, which is very close to the solar model age of the Sun (4.57 Gyr) given in Section 11.1.[16] It seems that the Allende meteorite is a nearly primordial remnant of the early solar nebula (as are other meteorites).

A chemical analysis of the samples revealed that the meteorite's composition is close to solar (similar to the Sun's photosphere), with some exceptions; the most volatile elements (H, He, C, N, O, Ne, and Ar) are underabundant, and lithium (Li) was found to be overabundant. The relative underabundance of volatiles can be understood by assuming that the Allende meteorite condensed out of the inner portion of the solar nebula where the temperature was too high for those elements to be included in solar concentrations.[17] Allende's lithium content is probably overabundant relative to the Sun because the Sun has actually destroyed much of its own complement of that element during the star's lifetime.

CAIs and Chondrules

Contained in the Allende samples are two types of nodules embedded in a **matrix** of dark silicate material. The **calcium- and aluminum-rich inclusions** (CAIs, also known as **refractory inclusions**) are small pockets of material ranging in size from microscopic to 10 cm in diameter that are relatively overabundant in calcium, aluminum, and titanium when compared with the remainder of the meteorite. This is significant because they are the

[16]Refer to Section 20.4 for a discussion of radioactive dating.

[17]Of course, light gases such as hydrogen and helium easily escape low-mass objects such as meteorites.

most refractory (least volatile) of the primary elements in meteoritic material. It seems that the CAIs have undergone repeated episodes of evaporation and condensation. **Chondrules** are spherical objects (1–5 mm across) made predominantly of SiO_2, MgO, and FeO, which seem to have cooled very rapidly from a molten state. Apparently no more than one melting and cooling event occurred for a given chondrule, and some chondrules may have been only partially molten.

A particularly intriguing discovery in the Allende CAIs is the overabundance of ${}^{26}_{12}\mathrm{Mg}$. Because this particular nuclide is produced by the radioactive decay of ${}^{26}_{13}\mathrm{Al}$ (recall Eq. 22.5), which is known to be produced by supernovae, the meteorite may have formed out of material significantly enriched with supernova ejecta. Moreover, because the half-life of ${}^{26}_{13}\mathrm{Al}$ is relatively short by astronomical timescales, the meteorite must have formed within a few million years or so following the production of the ${}^{26}_{13}\mathrm{Al}$. This suggests that a supernova shock wave may have triggered the collapse of the solar nebula. Because the material from the supernova should not be expected to mix thoroughly with the original nebula, regions of enhanced abundance would probably exist out of which objects such as the Allende meteorite could form. An alternative mechanism for the production of the required ${}^{26}_{13}\mathrm{Al}$ has also been proposed: Intense flares during pre-main-sequence T-Tauri and FU Orionis phases appear capable of synthesizing ${}^{26}_{13}\mathrm{Al}$. This mechanism seems to eliminate the need for a possibly ad hoc supernova trigger. More will be said about the formation and evolution of the Solar System in Section 23.2.

Carbonaceous and Ordinary Chondrites

The Allende meteorite is one example of a class of primitive specimens known as **carbonaceous chondrites**, so named because they are rich in organic compounds and contain chondrules. They may also include appreciable amounts of water in their silicate matrix. The matrix even records the existence of a fairly strong primordial magnetic field (about equal in strength to the value of Earth's present-day field). **Ordinary chondrites** contain fewer volatile materials than the carbonaceous chondrites, implying that they formed in a somewhat warmer environment. Both general types of chondrites are **chemically undifferentiated stony meteorites**.

Chemically Differentiated Meteorites

Several forms of **chemically differentiated meteorites** have also been discovered. Igneous stones, known as **achondrites**, do not contain any inclusions or chondrules; instead, they were formed entirely out of molten rock. **Iron** meteorites do not contain any stony (silicate) material, but they may be composed of up to 20% nickel. About three-quarters of all iron meteorites have long iron–nickel crystalline structures, up to several centimeters long and known as **Widmanstätten patterns**, that could have developed only if the crystal cooled very slowly over millions of years.[18] **Stony–iron** meteorites contain stony inclusions in a matrix of iron–nickel. **Stones** (chondrites and achondrites) make up about 96% of all the meteorites that hit Earth, irons account for about 3% of the total, and stony–irons make up the remainder (1%).

[18] The patterns were named for Count Alois von Widmanstätten, director of the Imperial Porcelain Works in Vienna, who discovered them in 1808.

Sources of Meteorites

The vast majority of all meteorites probably originate from asteroids, either chipped off their parents or liberated from the deep interior during a catastrophic collision. For a sufficiently large asteroid, significant gravitational separation may have occurred, as suggested by the M-type minor planets. The exposed metallic cores are the source of the irons, and the core–rock interface is the source of the stony–irons. Other asteroids underwent very little chemical alteration during their lives and may account for the chondrites.

The reflection spectra of asteroids can be compared with meteorite samples to test whether the asteroids could be the source of objects striking Earth. Figure 22.21 shows the strong correlations between the spectra of some asteroids and meteorites. Note, for instance, that the asteroid 176 Iduna has a spectrum very similar to that of the carbonaceous chondrite, Mighel, while the basaltic surface of 4 Vesta agrees well with that of the achondrite meteorite, Kapoeta.

An unusual achondrite was discovered on the ice cap of Antarctica in 1982.[19] It has the chemical makeup of rocks collected from the lunar highlands by the *Apollo* astronauts. Clearly this achondrite was ejected from the Moon instead of from an asteroid. Because the escape velocity of the Moon is much larger than the escape velocities of asteroids, the discovery was certainly unexpected. Even more surprising, a small handful of meteorites have been discovered whose ages date back only 1.3 Gyr. Because these stones are much younger than the surface of the Moon, they must have originated on a body that has been

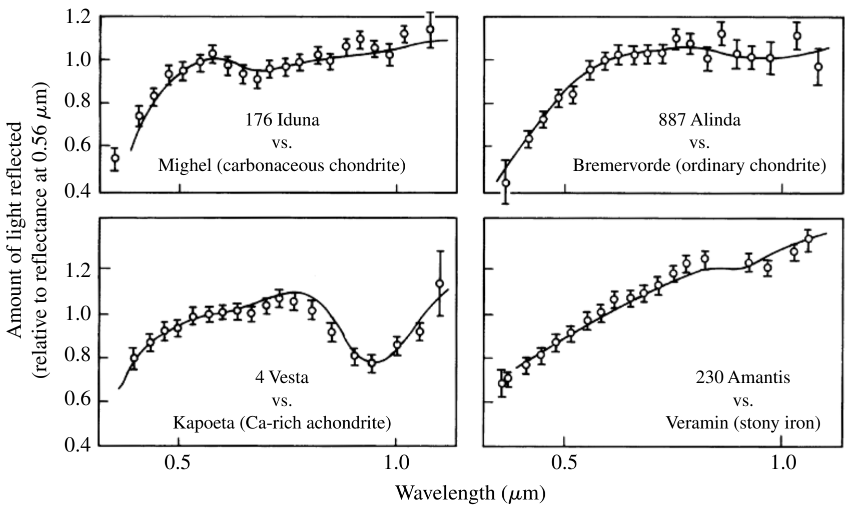


FIGURE 22.21 Comparisons between the infrared spectra of asteroids and meteorites. The reflection data for the asteroids are depicted by open circles with attached error bars. The laboratory spectra of the meteorites are given as solid curves. (Adapted from a figure courtesy of C. R. Chapman, in *The New Solar System*, Third Edition, Beatty and Chaikin (eds.), Cambridge University Press and Sky Publishing, Cambridge, MA, 1990.)

[19]Antarctica is an excellent site for finding meteorites. Any rock lying on the surface of a glacier is almost certainly extraterrestrial in origin.

TABLE 22.3 The Dates and Parent Bodies of Principal Meteor Showers.

Shower	Approximate Date	Parent Body
Quadrantid	January 3	(unknown)
Lyrid	April 21	Comet 1861 I
Eta Aquarid	May 4	Comet Halley
Delta Aquarid	July 30	(unknown)
Perseid	August 11	Comet Swift–Tuttle
Draconid	October 9	Comet Giacobini–Zinner
Orionid	October 20	Comet Halley
Taurid	October 31	Comet Encke
Andromedid	November 14	Comet Biela
Leonid	November 16	Comet 1866 I
Geminid	December 13	Asteroid 3200 Phaeton

geologically active more recently. The only real candidate is Mars, with an escape velocity of 5 km s^{-1}. At least one of the meteorites has inclusions of shock-melted glass that contain noble gases and nitrogen in the same proportions as the Martian atmosphere. However, also recall that Mars has produced at least one very old meteorite, ALH84001, discussed in Section 20.5 on page 764.

A number of **meteor showers** occur near the same dates every year, during which time meteors seem to emanate from a fixed position on the celestial sphere, known as a **radiant**. The source of the meteorites is debris left in the orbits of comets or asteroids that happen to intersect Earth's orbit. As Earth passes through the body's orbit, material rains down as if that material were coming from a position in the sky that Earth happens to be moving toward at the time; hence the radiant. Most parent bodies of meteor showers are comets, although at least one object, 3200 Phaeton, is classified as an asteroid. Meteor showers are named for the constellation in which their radiants lie. A list of the principal meteor showers, their approximate dates of maximum, and the parent object (if known) is given in Table 22.3.

A History of Collisions with Earth

By now it should be apparent that objects throughout the Solar System have been subjected to numerous and sometimes violent collisions; Earth is no exception. Even as recently as 50,000 years ago, an iron meteorite, estimated to be 50 m in diameter, hit the ground in Arizona, producing a crater 1.2 km wide and 200 m deep (Fig. 22.22). There is also strong evidence to support the hypothesis that a stony asteroid exploded in the atmosphere above Siberia in 1908 (an episode known as the Tunguska event). The detonation leveled trees in a radial pattern for 15 km in every direction. It is even reported that the blast wave knocked a man off his porch 60 km from the epicenter and that the explosion was audible at distances of up to 1000 km. Estimates place the energy released during the Tunguska event at 5×10^{17} J, equivalent to a nuclear explosion of 12 MTons.

Could other, even more energetic collisions have produced catastrophic consequences for life on Earth at the time? In about 1950, Ralph Baldwin suggested that meteoritic impacts

FIGURE 22.22 The 50,000-year-old Meteor Crater (also known as Barringer's Crater) in Arizona is 1.2 km in diameter and 200 m deep. It was produced by an iron meteorite estimated to be 50 m in diameter. (Courtesy of D. J. Roddy and K. Zeller, USGS.)

could have been responsible for the mass extinctions of many species seen in the paleontological record. Support for this hypothesis came in 1979 when geologist Walter Alvarez and his father, Luis Alvarez (1911–1988), a Nobel Prize winner in physics, announced the discovery of high abundances of iridium in a dark-colored clay that was located in the geologic strata at the Cretaceous–Tertiary boundary (commonly called the K–T boundary). The K–T boundary corresponds in time to the extinction, 65 million years ago, of 70% of the species then in existence, including the last of the dinosaurs. Since their original discovery in the Appenine Mountains of Italy, the anomalously high iridium concentrations have been seen throughout the world at the K–T boundary.

The significance of iridium is that it is rare in rock found near Earth's surface. This is because iridium is readily soluble in molten iron (it is a **siderophile**) and as such participated in the chemical differentiation of heavy elements sinking toward the core of Earth. However, iridium is fairly common in iron-rich meteorites. The amount of iridium present in the K–T clay strata, where it is thousands of times more abundant than is typical of ordinary rock, is consistent with an impact by a stony asteroid that measured 6 to 10 km across (or perhaps the object was a slightly larger comet).[20] An impactor of this size should have produced a crater some 100 to 200 km in diameter.

Shocked mineral grains have also been found worldwide at the K–T boundary but are most abundant in North America, suggesting that the impact (or impacts) could have occurred there. Attention has focused on an ancient impact site along the northern coast of the Yucatan peninsula, near the town of Chicxulub. A nearly semicircular structure at least 180 km across is located there and, based on radioactive dating, appears to be of the correct age.[21] Based on the size of the crater, the energy of the impact is estimated to have been 4×10^{22} J, the equivalent of 10^{13} tons of TNT. An impact of that magnitude at the Chicxulub site could also account for evidence of an enormous tidal wave (a **tsunami**) that apparently traveled as far north as central Texas.

How could such an ocean impact have led to mass extinctions? If a meteorite of the size suggested hit in the ocean, it would vaporize a large amount of water. Some of this

[20]For comparison, the vertical rise of a typical mountain in the Rockies is about 1.5 km above the valley floor, and ocean depths are approximately 6 km.

[21]Some scientists have suggested that the diameter of the crater may be more like 300 km.

water would wash out airborne dust while the rest of the moisture would increase the greenhouse effect. As the temperature rose, even more water would evaporate into the atmosphere. Global atmospheric and ocean surface temperatures could rise by as much as 10 K through this enhanced greenhouse effect. (Recall the discussion on page 743 of the runaway greenhouse effect on Venus.)

Alternatively, if a major impact were to occur on land, a tremendous amount of dust would be injected into the atmosphere. As a consequence, the albedo would increase and more solar radiation would be reflected back into space, cooling the surface.[22]

In either case, as the meteorite passed through the atmosphere, the enormous amount of kinetic energy available in the impactor would have produced searing heat and generated devastating fires. It would also have reacted with appreciable amounts of nitrogen, producing nitrogen oxides and nitric acid. The ensuing acid rain would have damaged delicate land-based and aquatic ecosystems, killing vegetation and destroying much of the remaining food source. Carbon soot is found in the K–T clay layer, and there is also geologic evidence suggesting that flowering plants were destroyed in some regions for periods of at least several thousand years. Regardless of whether the impact occurred on land or in an ocean, the global environment would have been dramatically affected.

Even if asteroids or comets did not kill the dinosaurs and other creatures, there is clear evidence that major impacts have occurred in the past. By some estimates, the probability of the occurrence, during our lifetimes, of a cataclysmic impact that would be capable of destroying civilizations is perhaps as high as one in a few thousand. In light of this rather surprising statistic, some scientists have suggested that we should build a global asteroid–comet defense system. Although no definite plans have yet been formulated, conferences have been held to discuss the possibility.

The Basic Building Blocks of Life

Ironically, even though impactors have been proposed as the mass murderers of some life forms on Earth, a number of carbonaceous chondrites have been found to contain many of the basic building blocks of life. Seventy-four amino acids have been found in one meteorite alone (the Murchison meteorite, which fell in Australia in 1972). Of those, seventeen are important in terrestrial biology. In addition to the amino acids, all four of the bases that cross-link the double helix of the DNA molecule (guanine, adenine, cytosine, and thymine), and the fifth base that is important in cross-linking in RNA (uracil), have been discovered in the Murchison meteorite. Other molecules important to life on Earth (such as fatty acids) have also been found in carbonaceous chondrites. Of course, it is a long way from producing relatively simple amino acids and cross-linking bases to the generation of the extremely complex DNA and RNA molecules, but these discoveries indicate that the fundamental chemistry necessary to start the process can occur in an extraterrestrial environment.

[22]It has been suggested that such a situation could also arise following a large-scale nuclear war. This scenario has been referred to as "nuclear winter."

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PROBLEMS

22.1 **(a)** Assume that a spherical dust grain located 1 AU from the Sun has a radius of 100 nm and a density of 3000 kg m^{-3}. In the absence of gravity, estimate the acceleration of that grain due to radiation pressure. Assume that the solar radiation is completely absorbed.

(b) What is the gravitational acceleration on the grain?

22.2 In Problem 21.16, the Poynting–Robertson effect was shown to be important in understanding the dynamics of ring systems. The Poynting–Robertson effect, together with radiation pressure, is also important in clearing the Solar System of dust left behind by comets and colliding asteroids (the dust that is responsible for the zodiacal light).

(a) Beginning with Eq. (21.6), found in Problem 21.15, show that the time required for a spherical particle of radius R and density ρ to spiral into the Sun from an initial orbital radius of $r \gg R_\odot$ is given by Eq. (22.4). Assume that the orbit of the dust grain is approximately circular at all times.

(b) Find the radius of the largest spherical particle that could have spiraled into the Sun from the orbit of Mars during the Solar System's 4.57-billion-year history. Take the density of the dust grain to be 3000 kg m^{-3}.

22.3 Estimate the amount of mass lost by Comet Halley during its most recent trip through the inner Solar System. Take into consideration the fact that the comet exhibits significant activity only during a short period of time near perihelion (an interval of approximately one year). Compare your answer with the total amount of mass present in the nucleus. Assuming that the mass loss rates are the same for each trip, how many more trips might the comet be able to make before it becomes extinct?

22.4 In the text it was mentioned that *nongravitational perturbations* were used to estimate the mass of Comet Halley. How might this be done?

22.5 Comet 1943 I, which last passed through perihelion on February 27, 1991, has an orbital period of 512 years and an orbital eccentricity of 0.999914. This is one member of the class of Sun-grazing comets.

(a) What is the comet's semimajor axis?

(b) Determine its perihelion and aphelion distances from the Sun.

(c) What is the most likely source of this object, the Oort cloud or the Kuiper belt?

22.6 Using Kepler's laws, verify that the 2:1 and 3:1 orbital resonances of Jupiter correspond to the two prominent Kirkwood gaps indicated in Fig. 22.13.

22.7 Vesta orbits the Sun at a distance 2.362 AU and has an albedo of 0.38 (unusually reflective for an asteroid).

(a) Estimate Vesta's blackbody temperature, assuming that the temperature is uniform across the asteroid's surface.

(b) If Vesta's radius is 250 km, how much energy does it radiate from its surface every second?

22.8 Figure 22.14 makes it appear that the asteroid belt is saturated with objects. In this problem we will consider the fraction of the volume actually occupied by asteroids.

(a) If there are 300,000 large asteroids between 2 AU and 3 AU from the Sun, and each asteroid is assumed to be spherical with a radius of 100 km, determine the total volume occupied by the asteroids considered here.

(b) Model the region in which these asteroids orbit as an annulus with an inner radius of 2 AU, an outer radius of 3 AU, and a thickness of 2 $R_\odot$. Determine the volume of the region.

(c) What is the ratio of the volume occupied by asteroids to the volume of the region in which they orbit?

(d) Comment on the validity of a spaceship needing to maneuver quickly through a dense population of asteroids as frequently depicted in popular science fiction movies.

22.9 In this problem you will estimate the amount of energy released per second by the radioactive decay of $^{26}_{13}\mathrm{Al}$ inside Vesta during its lifetime.

(a) Vesta has a radius of 250 km and a density of 2900 kg m^{-3}. Assuming spherical symmetry, estimate the asteroid's mass.

(b) Assume for the moment that the asteroid is composed entirely of silicon atoms. Estimate the total number of atoms inside Vesta. The mass of one silicon atom is approximately 28 u.

(c) The mass of $^{26}_{13}\mathrm{Al}$ is 25.986892 u and the mass of $^{26}_{12}\mathrm{Mg}$ is 25.982594 u. How much energy is released in the decay of one aluminum atom? Express your answer in joules.

(d) The ratio of $^{26}_{13}\mathrm{Al}$ to all aluminum atoms formed in a supernova is about 5×10^{-5}, and aluminum constitutes approximately 8680 ppm (parts per million) of the atoms in a chondritic meteorite. Assuming that these values apply to Vesta, estimate the number of $^{26}_{13}\mathrm{Al}$ atoms originally present in the asteroid.

(e) Find an expression for the amount of energy released per second in the decay of $^{26}_{13}\mathrm{Al}$ within Vesta as a function of time, and plot your results over the first 5×10^7 years on semilog graph paper. You may find Eq. (15.9) useful.

(f) How much time was required after the formation of Vesta before energy production due to the radioactive decay of $^{26}_{13}\mathrm{Al}$ dropped to 1×10^{13} W, comparable to the current rate of energy output from the asteroid? See Problem 22.7.

22.10 With the aid of a diagram, explain why it is best to observe a meteor shower between 2 A.M. and dawn instead of in the early evening. *Hint:* Consider the velocities of the infalling meteors and the orbital and rotational motions of Earth.

22.11 Suppose that the Tunguska event was caused by an asteroid colliding with Earth. Assume that the density of the object was 2000 kg m^{-3} and that it exploded above the surface of the planet traveling at a rate equal to Earth's escape velocity. If all of the energy of the explosion was derived from the asteroid's kinetic energy, estimate the mass and radius of the impacting body (assume spherical symmetry).

CHAPTER

23 Formation of Planetary Systems

23.1 *Characteristics of Extrasolar Planetary Systems*
23.2 *Planetary System Formation and Evolution*

23.1 ■ CHARACTERISTICS OF EXTRASOLAR PLANETARY SYSTEMS

You will recall that in Section 7.4 we discussed several methods that have been used to detect extrasolar planets (sometimes referred to as *exoplanets*). With the rapid increase in the number of known planets beyond our own Solar System, important new information is being gathered concerning how planetary systems form and evolve. In addition to these systems being interesting to study in their own right, this increase in knowledge about extrasolar planets helps to inform us about our own Solar System.

Detections through the Reflex Radial Velocity Technique

As mentioned in Section 7.4, the most effective method of discovering extrasolar planets to date has been through the measurement of the reflex radial velocity of the parent star. With the exception of the pulsar, PSR 1257+12, 51 Pegasi was the first star (other than our Sun) found to have a planet in orbit around it. Michel Mayor and Didier Queloz of the Geneva Observatory made the announcement in October 1995 of a planet with a period of $P = 4.23077$ d in a nearly circular orbit ($e < 0.01$) around 51 Peg (a more recent radial velocity curve of 51 Peg obtained by Geoffrey Marcy and his collaborators is shown in Fig. 23.1). Since the system is not eclipsing, and the planet is too faint to be visually identified, the inclination of the orbit of the planet (i) is unknown. As a result, only the quantity $m \sin i$ can be determined for the planet from radial velocity measurements (see, for example, Eq. 7.5). Given that the parent star is a near twin of our Sun, with a spectral classification of G2V–G3V, implying a stellar mass of approximately 1 $M_\odot$, the lower mass limit of the orbiting planet is obtained from the maximum radial velocity wobble of the star.

Example 23.1.1. To determine the minimum mass of the planet orbiting 51 Peg, we must first determine its orbital velocity. From Kepler's third law (Eq. 2.37), and assuming that the mass of the star is $m_{51} = 1\ M_\odot$ and that the planet's mass, m, is insignificant ($m \ll m_{51}$), we find

$$a = \left[\frac{GP^2(m_{51} + m)}{4\pi^2}\right]^{1/3} = 7.65 \times 10^9 \text{ m} = 0.051 \text{ AU}.$$

Since the orbit of the planet is nearly circular, the orbital speed of the planet is

$$v = 2\pi a/P = 131 \text{ km s}^{-1}.$$

Employing Eq. (7.5), and noting from Fig. 23.1 that the amplitude of the star's observed radial velocity is $v_{r,\max} = v_{51} \sin i = 56.04 \text{ m s}^{-1}$, we find that

$$m \sin i = m_{51} \frac{v_{51} \sin i}{v} = 8.48 \times 10^{26} \text{ kg} = 0.45 \text{ M}_\text{J},$$

where M_J is the mass of Jupiter. Since $\sin i \leq 1$, the mass of the planet, 51 Peg b, must be greater than 0.45 M_J.

51 Peg b is one example of a "hot Jupiter," one of a number of extrasolar planets that have been discovered having Jupiter-class masses but orbiting very close to their parent star.

Multi-Planet Systems

A number of extrasolar planetary systems have been found through the radial velocity technique to have multiple planets in orbit about the central stars. An example of one such system is υ Andromedae; see Fig. 23.2. After the orbital perturbations due to the 4.6-d orbit of one planet were removed from the radial velocity curve of the star, evidence remained of additional perturbations. The υ And system contains at least three planets with orbital periods of 4.6 d, 241 d, and 1284 d, with $m \sin i$'s of 0.69 M_J, 1.89 M_J, and 3.75 M_J, respectively. The mass of the F8V parent star is estimated to be $1.3 \text{ M}_\odot$ (Appendix G).

As of May 2006, 193 extrasolar planets have been detected in 165 planetary systems. While most of the planetary systems have had just one planet detected in them so far, 20 systems are known to be multi-planet systems.

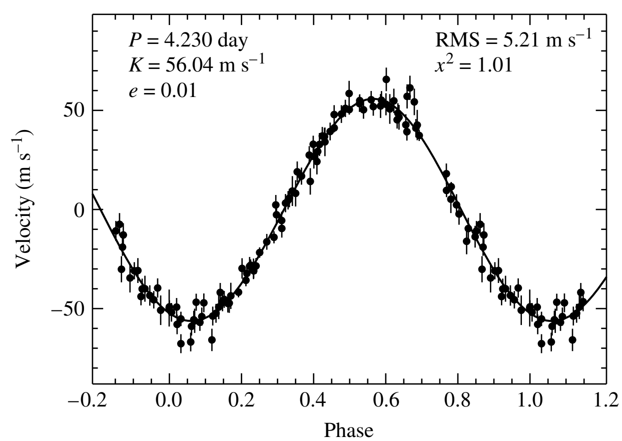

FIGURE 23.1 The radial velocity measurements of 51 Pegasi, revealing the presence of a planet orbiting only 0.051 AU from the star. The sinusoidal shape of the velocity curve is evidence of a very low orbital eccentricity; recall the discussion in Section 7.3. (Figure adapted from Marcy, et al., *Ap. J.*, *481*, 926, 1997.)

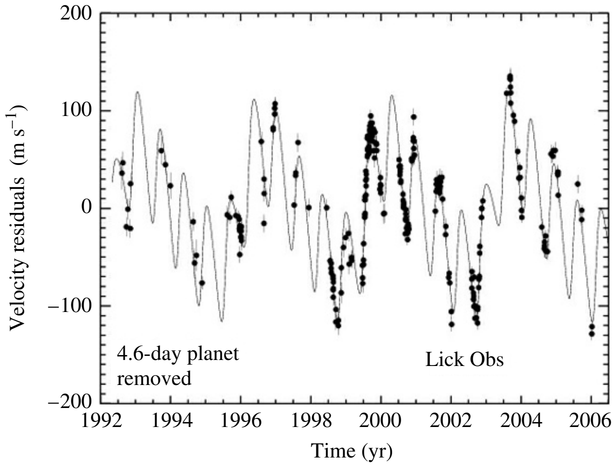

FIGURE 23.2 The residuals in the radial velocity measurements of υ Andromedae after the gravitational perturbations of the 4.6-day planet have been removed. The data reveal the presence of at least three planets orbiting υ And. (Adapted from a figure provided by Debra A. Fischer, private communication.)

The Mass Distribution of Extrasolar Planets

Initially, the radial velocity technique was able to discover only very massive (Jupiter-class) planets in close-in orbits around their parent stars. One of the reasons for this selection effect is that these objects exert the greatest gravitational influence on their parent star and generate the largest reflex radial velocities. The other reason is that a star must be observed over a time interval greater than the orbital period of the planet before the existence of the planet can be confirmed. As the amount of time increases for the systems being surveyed, the longer time-line data have allowed researchers to find lower-mass planets and planets orbiting farther from the star. The lowest-mass planet discovered to date is in a multiple system orbiting Gliese 876 and has an $m \sin i = 0.023\ \mathrm{M_J}$, which is just $7.3\ \mathrm{M_\oplus}$. The largest orbit detected thus far using the reflex motion technique is in the multiple system 55 Cancri, with a semimajor axis of 5.257 AU and an orbital period of 4517 d = 12.37 yr.

Over time, this selection effect is systematically diminishing. As is evident from statistical studies of the systems investigated so far, nature seems able to produce planets with a range of masses, with the lowest-mass planets being the most common. When binned by mass interval (see Fig. 23.3), the number of planets in each mass bin varies as

$$\frac{dN}{dM} \propto M^{-1}. \tag{23.1}$$

The Distribution of Orbital Eccentricities

It is also interesting to note the relationship between orbital eccentricity (e) and semimajor axis for extrasolar planets (Fig. 23.4). Those planets that are orbiting close to their parent star

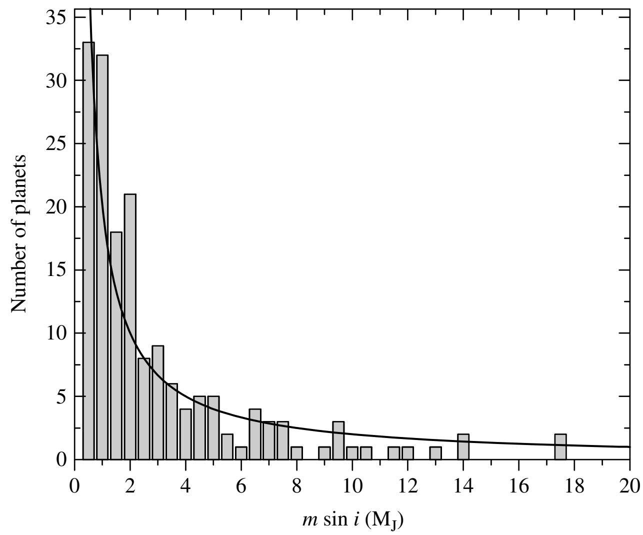

FIGURE 23.3 The number of planets in mass bins of interval 0.5 M_J. The solid line is given by Eq. (23.1). (Data from *The Extrasolar Planets Encyclopedia*, `http://exoplanet.eu`, maintained by Jean Schneider.)

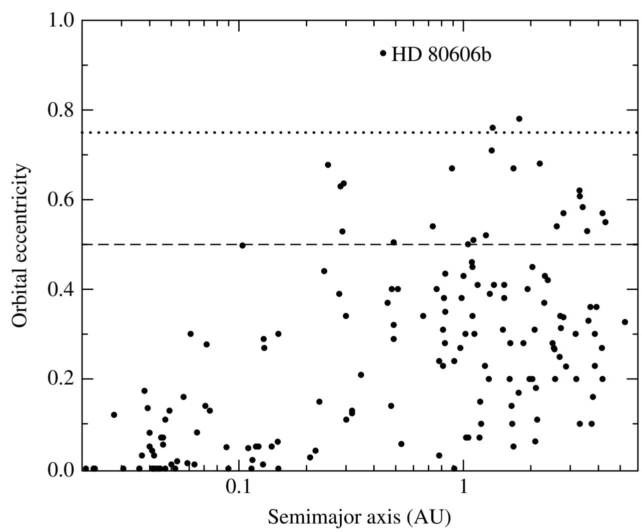

FIGURE 23.4 The orbital eccentricities of the known extrasolar planets as a function of semimajor axis. Only 3 planets ($< 2\%$) are known to have eccentricities greater than 0.75, and less than 15% have eccentricities greater than 0.5. (Data from *The Extrasolar Planets Encyclopedia*, `http://exoplanet.eu`, maintained by Jean Schneider.)

tend to have circularized orbits (or at least orbits with smaller eccentricities). Planets orbiting farther from their parent star may have high orbital eccentricities, with the maximum value determined to date being a planet orbiting HD 80606 with $e = 0.927$ and a semimajor axis of 0.439 AU. However, from the data obtained thus far, only 15% of planets are known to have eccentricities of greater than 0.5, and less than 2% have eccentricities in excess of 0.75.

With the small number of high-eccentricity planets, it is important to ask whether or not there is something unique about the systems in which they are found. HD 80606 turns out to be one member of a wide stellar binary system, the other member being HD 80607. These two G5V stars are nearly identical and slightly smaller than the Sun. The two stars are also separated by a projected distance of 2000 AU. It has been suggested that the gravitational perturbations exerted on the planet, HD 80606b, by HD 80607 may have pumped its orbit up to its current very high eccentricity. In support of this suggestion, another planet with a high eccentricity ($e = 0.67$), 16 Cyg Bb, is also a member of a binary star system. However, the timescale for the gravitational perturbations provided by HD 80607 that would cause the orbital eccentricity of HD 80606b to significantly increase is estimated to be 1 Gyr. This long period of time comes from the necessary resonant alignment of the second star with the planet. The 1-Gyr timescale must be compared against the 1-Myr timescale provided by the general relativistic effect of the advance of the periastron of HD 80606b's orbit due to its parent star (recall the discussion of the advance of perihelion of Mercury's orbit in Section 17.1). It is argued that the general relativistic effects would completely overwhelm the perturbations from HD 80607 unless there were a third body in the HD 80606 system with an orbital period of roughly 100 yr that could also gravitationally influence HD 80606b. So far, a third object has not been discovered.

Two conclusions may be drawn from these data: (1) Planets with orbital periods of less than 5 days tend to have the smallest eccentricities ($e < 0.17$, with 80% of those having $e < 0.1$), probably due to strong tidal interactions with the parent star, and (2) planets sufficiently far from the parent star may have fairly large orbital eccentricities, but typically less than about 0.5. It seems that our own Solar System is somewhat unique, at least compared to the systems studied to date, in that our planets tend to have orbital eccentricities that are very small (excluding the Kuiper belt objects).

The Trend toward High Metallicity

An additional important trend has also been emerging from the extrasolar planetary system data obtained to date. It appears that there is a strong tendency for planetary systems to preferentially form around metal-rich (Population I) stars. One way to quantify the **metallicity** is by comparing the ratios of iron to hydrogen in stars relative to our Sun, defining the metallicity to be

$$\boxed{[\mathrm{Fe/H}] \equiv \log_{10}\left[\frac{(N_{\mathrm{Fe}}/N_{\mathrm{H}})_{\mathrm{star}}}{(N_{\mathrm{Fe}}/N_{\mathrm{H}})_{\odot}}\right],} \tag{23.2}$$

where N_{Fe} and N_{H} represent the *number* of iron and hydrogen atoms, respectively. Stars with $[\mathrm{Fe/H}] < 0$ are metal-poor relative to the Sun, and stars with $[\mathrm{Fe/H}] > 0$ are relatively

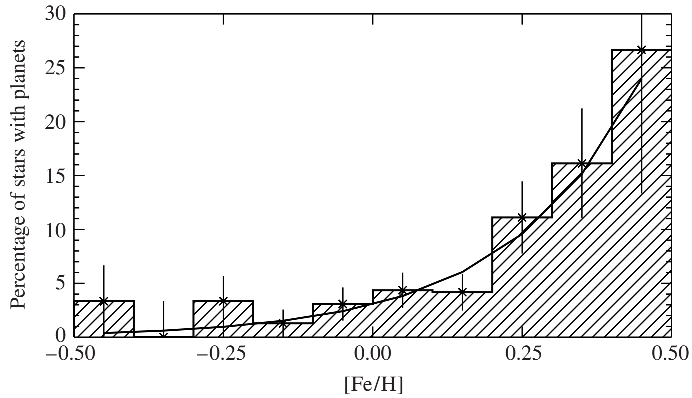

FIGURE 23.5 The percentage of stars found to have planetary systems, relative to the number of stars investigated in each metallicity bin. The solid curve is given by Eq. (23.3). (Figure adapted from Fischer and Valenti, *Ap. J.*, *622*, 1102, 2005.)

metal-rich. For comparison, extremely metal-poor (Population II) stars in the Milky Way Galaxy have been measured with values of [Fe/H] as low as -5.4, while the highest values for metal-rich stars are about 0.6.

As can be seen in Fig. 23.5, most stars with planetary systems detected so far tend to be metal-rich compared to the Sun. Those stars that do have a metallicity lower than the solar value are only moderately lower. The data in Fig. 23.5 are plotted as the percentage of stars in a given metallicity bin that were well studied and found to have planetary systems. According to the sample of 1040 F, G, and K stars used in the study, the data seem to be well-fit by the relationship

$$\mathcal{P} = 0.03 \times 10^{2.0[\mathrm{Fe/H}]}, \tag{23.3}$$

where $\mathcal{P}$ is the probability of a star having a detectable planetary system.

Measuring Radii and Densities Using Transits

The transit of a planet across the disk of the parent star provides further information about the planet (recall Fig. 7.12). From the timing of the eclipse, and using atmospheric models of the star that include limb darkening, it is possible to determine the planet's radius. Of course, once the radius is determined, the planet's average density may also be computed. From the small number of systems where this has been possible, it appears that the Jupiter-class planets have densities that are similar to those of the gas giants in our Solar System (see Fig. 23.6). However, some of the so-called "hot Jupiters" that orbit close to the parent star appear to be somewhat inflated (e.g., HD 209458b and OGLE-TR-10b). The simple answer to explain the effect, namely the higher surface temperature due to the planet's proximity to the parent star, doesn't seem to explain all of the close-in systems, so apparently another source (or sources) of heat is required to puff up the planets. Some of the suggestions for solving the puzzle include tidal dissipation due to ongoing circularization of the orbit (perhaps involving another undetected object trying to simultaneously pump up the orbital

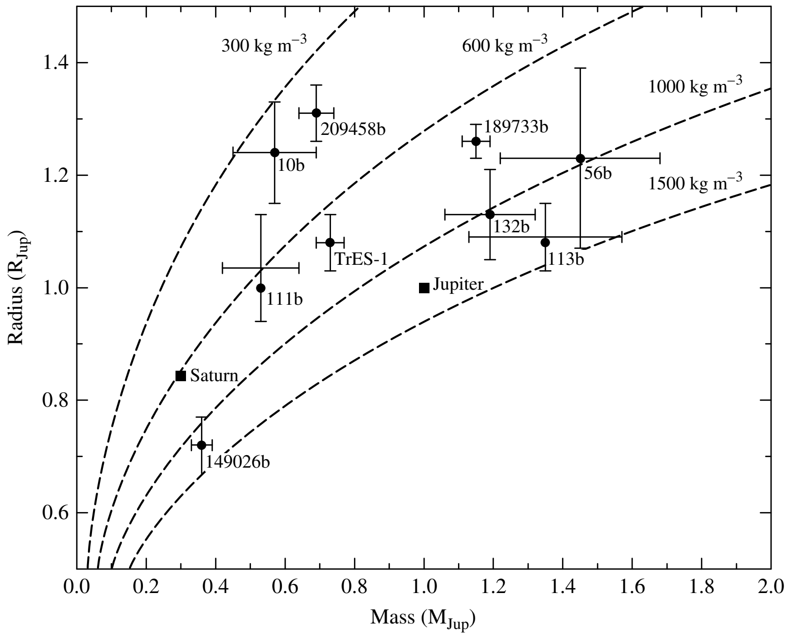

FIGURE 23.6 The relationship between radius and mass for transiting extrasolar planets. The dashed lines correspond to specific average mass densities. (Adapted from a figure provided by Debra A. Fischer, private communication.)

eccentricity), a misalignment between the planet's orbital plane and the equator of the star, and dissipation of atmospheric currents in the planet as its gas migrates from the hot, substellar point to the cooler region on the back of the planet (Hadley circulation).

At least one planet appears to have a massive rocky core. The G0IV star HD 149026 has a transiting "hot Saturn," with an $m \sin i = 0.36\ \mathrm{M_J}$. From the transits, the orbital inclination has been determined to be $85.3° \pm 1.0°$, allowing a determination of the mass of the planet (not just a lower limit) of $0.36\ \mathrm{M_J} = 1.2\ \mathrm{M_S}$, where $\mathrm{M_S}$ designates the mass of Saturn. The timing of the transits also yields a radius of $0.725 \pm 0.05\ \mathrm{R_J}$ for the planet, implying an average density of 1253 kg m^{-3}, which is 94% the density of Jupiter but 1.8 times the density of Saturn. The star itself has a mass and radius of $1.3 \pm 0.1\ \mathrm{M_\odot}$ and $1.45\ \mathrm{R_\odot}$, respectively. In addition, the star's metallicity is $[\mathrm{Fe/H}] = 0.36$, making it a significantly metal-rich star. Based on computer models of the planet's interior, it appears that the planet possesses a 67 $\mathrm{M_\oplus}$ core composed of elements heavier than hydrogen and helium, assuming that the core density is 10,500 kg m^{-3}, which is believed to be similar to Saturn's core. If the core density is only 5500 kg m^{-3}, then the calculated core mass would be even larger (78 $\mathrm{M_\oplus}$).

The Detection of an Extrasolar Planet Atmosphere

Transits of extrasolar planets across the disks of their parent stars also provide for the possibility of detecting extrasolar planetary atmospheres. The first planet for which this was accomplished was HD 209458b. David Charbonneau and his collaborators were able to detect the spectroscopic signature of sodium at its resonance doublet wavelength of 589.3 nm by noting differences in the spectrum of the star as the planet passed in front of it. The starlight passing through the planet's atmosphere produced an enhanced absorption feature at that wavelength. The effect was very subtle, with deepening of the absorption feature by only an additional factor of 2.32 ± 10^{-4} relative to adjacent wavelength bands during the transit. It has been proposed that the spectral signatures of water, methane, and carbon monoxide may be able to be detected in this way as well.

Distinguishing Extrasolar Planets from Brown Dwarfs

With the detection of a few extrasolar planets having masses more than a factor of ten larger than the mass of Jupiter, the question is again raised concerning the definition of a planet. At the low-mass end, large Kuiper belt objects such as Pluto have been classified as planets. At the upper end, what distinguishes a planet from a brown dwarf?

Two different criteria have been proposed to answer this question. One suggestion is tied to the formation process of planets and stars. As we discussed in Chapter 12, stars form from the gravitational collapse of a gas cloud. As we will explore further in the next section, planets are generally believed to form from a bottom-up accretion process, although there has been speculation that gravitational collapse in the star's accretion disk may also produce planets. One proposed definition of planet is that it is an object that forms through a process beginning with the bottom-up accretion of planetesimals, whereas a brown dwarf forms directly from gravitational collapse. The challenge with such a definition is determining after the fact how a particular object may have formed.

A second criterion that has been proposed is based on whether or not the object that forms is massive enough ever to have had nuclear fusion occur in its core. Computer models of very low-mass objects indicate that if the mass of the object is greater than 13 M_J, deuterium can burn while the object is forming. The rate of energy production would not be sufficient to stabilize the object during gravitational collapse, but deuterium burning can be sufficient to affect the luminosity of the object during collapse. At the other end, as mentioned in Section 10.6, stars with mass of at least 0.072 $M_\odot$ (75 M_J) for solar composition undergo nuclear fusion at a sufficient rate to stabilize them at the low-mass end of the main sequence. Thus, it is proposed that brown dwarfs should be considered as being those objects having masses between these two limits ($13\ M_J < M_{bd} < 75\ M_J$); in other words, brown dwarfs are "stars" that burn some deuterium but never reach a stable nuclear-burning phase during contraction. Given the difficulty with the formation-mechanism criterion, the nuclear-reaction/mass-based criterion is generally favored.

An Image of an Extrasolar Planet

In 2004 the first image of an extrasolar planet was obtained by Gael Chauvin and collaborators, using the European Southern Observatory's Very Large Telescope with an infrared detector; see Fig. 23.7. The parent star is a 25 M_J brown dwarf of spectral type M8.5, known as 2MASSWJ1207334−393254, or 2M1207 for short! The system was also resolved later by the Hubble Space Telescope's NICMOS instrument. The planet resides 55 AU from the brown dwarf and has an estimated mass of $5 \pm 2\ M_J$. From the infrared observations, the spectral type of the planet is between L5 and L9.5.

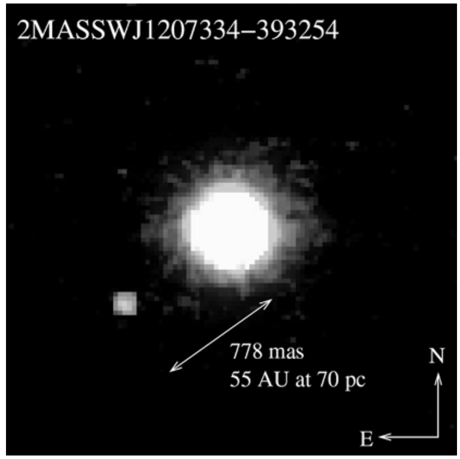

FIGURE 23.7 The first image obtained of an extrasolar planet. The planet is orbiting the brown dwarf 2MASSWJ1207334−393254. (Image courtesy of the European Southern Observatory.)

Future Space-Based Planet Searches

Given the dramatic success since the mid 1990s in detecting planetary companions of main-sequence stars, a number of projects are planned to further the search using space-based observatories:

- **COROT** (COnvection, ROtation, and planetary Transits) is a joint mission of France, ESA, Germany, Spain, Belgium, and Brazil that is designed to study stellar seismology and search for planetary transits. COROT is scheduled for launch in 2006.

- NASA's **Kepler** mission is slated for launch in 2008 and will search for transits of Earth-sized planets across their parent stars' disks. Specifically, the Kepler mission hopes to identify Earth-like planets in the habitable zone around solar-type stars out to a distance of about 1 kpc.

- The **SIM PlanetQuest** mission, scheduled for launch in 2011, is designed to obtain high-precision astrometric data (see Section 6.5). One of SIM's primary missions is to search for nearby extrasolar Earth-sized planets.

- The data obtained from Kepler and SIM will provide input data for another NASA mission, known as the **Terrestrial Planet Finder** (TPF). TPF, as it is currently envisioned, will be made of two complementary component missions: a *visible-light coronagraph*, scheduled for launch around 2014, and an *infrared nulling interferometer* that will be composed of five individual spacecraft flying in precise formation (to be launched before 2020). Together, the two components of the TPF should be able to identify Earth-like planets and measure their atmospheric chemistries. One goal of TPF is to try to detect the signatures of life in the atmospheres of other Earth-like planets.

- Sometime in 2015 or later, ESA plans to launch **Darwin**, a free-flying array of six infrared telescopes that will also act as an infrared nulling interferometer.

With the great focus on planetary searches currently under way from the ground and from space, and with additional space-based missions planned for the future, the tremendous advances in this field of modern astrophysics can be expected to continue.

23.2 ■ PLANETARY SYSTEM FORMATION AND EVOLUTION

The question of how Earth and the Solar System formed has intrigued humans in all cultures for thousands of years. In 1778 Georges-Louis Leclerc, Comte de Buffon (1707–1788) proposed that a giant comet collided with the Sun, causing the ejection of a disk of material that ultimately condensed to form the planets. Competing tidal theories argued that a close encounter with a passing star ripped material from the Sun. Unfortunately, each of these theories suffers from a number of difficulties, including inadequate energy, composition differences between the planets and the Sun, and the sheer improbability of such an event. Another class of theories suggested that the Sun accreted planetary material from interstellar space, taking care of the difficulty of composition differences between the Sun and the planets, but not those among the planets themselves. Yet another class of theories, the basis of today's models, argue for the simultaneous formation of the Sun and the planets out of the same nebula. Among the early proponents of these so-called nebular theories were René Descartes (1596–1650), Immanuel Kant (1724–1804), and Pierre-Simon, Marquis de Laplace (1749–1827).

Although a significant number of problems remain to be solved, there is now some sense of convergence on the basic components of planetary system formation. Throughout Part III (as well as in the rest of the book to this point), we have presented clues related to critical features of a comprehensive model, some obvious and others more subtle. Before discussing our present understanding of the formation of planetary systems, we will review some of these clues and the questions they raise.

Accretion Disks and Debris Disks

In Chapter 12 we introduced the wide range of observational data related to the formation and pre-main-sequence evolution of stars. It is clear from both observational and theoretical studies that stars form from the gravitational collapse of clouds of gas and dust. If a collapsing cloud contains any angular momentum at all (which it surely will), the collapse leads to the formation of an accretion disk around the growing protostar, as explored in Problem 12.18.

As a direct observational consequence of the conservation of angular momentum, numerous examples of accretion disk formation have been discovered and studied in detail, including the many proplyds observed in the Orion Nebula and elsewhere (Fig. 12.23) and the jets and Herbig–Haro objects associated with young protostars (Figs. 12.18 and 12.19). In addition, there is growing evidence that clumps of material exist in these disks.

There is also substantial evidence of **debris disks** around older stars, such as β Pictoris (Fig. 12.21). The implication is that material is left over in the disk after the star has finished forming. Debris disks may be the extrasolar analogs to the asteroid belt and the Kuiper belt.

Angular Momentum Distribution in the Solar System

However, one problem that has frustrated most attempts to put together an adequate picture of how our own Solar System developed concerns the present-day *distribution* of its angular momentum. In Problem 2.6 a simple calculation of the angular momentum in the Sun and Jupiter revealed that the orbital angular momentum of that planet exceeds the rotational angular momentum of the Sun by roughly a factor of twenty. A more detailed analysis shows that even though the Sun contains 99.9% of the mass, it contains only about 1% of the angular momentum of the entire Solar System, and most of the remainder is associated with Jupiter.[1] To complicate matters further, the Sun's spin axis is tilted 7° with respect to the average angular momentum vector of the planets, making it hard to envision how such a distribution of angular momentum could develop.

An additional interesting component of the angular momentum question concerns the amount of angular momentum possessed by other stars. It turns out that, on average, main-sequence stars that are more massive rotate much more rapidly and contain more angular momentum per unit mass than do less massive ones. Moreover, as can be seen in Fig. 23.8, a very discernible break occurs in the amount of angular momentum per unit mass as a function of mass near spectral class A5. If the total angular momentum of the Solar System were included, rather than just the angular momentum of the Sun, the trend along the upper

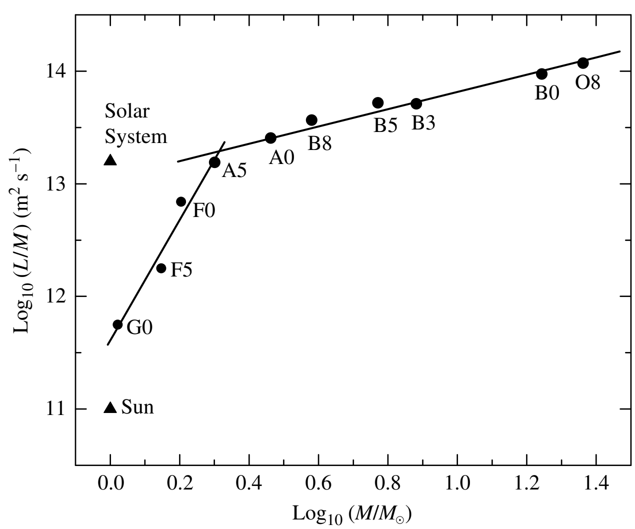

FIGURE 23.8 The average amount of angular momentum per unit mass as a function of mass for stars on the main sequence. The Sun's value and the total for the entire Solar System are indicated by triangles. Best-fit straight lines have been indicated for stars A5 and earlier, as well as for stars A5 and later (not including the Sun).

[1]The difference between the result obtained in Problem 2.6 and the one quoted here lies in the assumption made in Problem 2.6 that the Sun is a solid, constant-density sphere having a moment of inertia of $0.4M_\odot R_\odot^2$. In fact, the Sun does not rotate as a rigid body, and because it is centrally condensed, its moment of inertia is closer to $0.073M_\odot R_\odot^2$.

end of the main sequence would extend to include our Solar System as well (recall that the Sun is a G2 star).

A portion of the angular momentum problem may be solved by the transport of angular momentum outward via plasma drag in a corotating magnetic field. Charged particles trapped in the protosun's field would have been dragged along as the field swept through space. In response, the protosun's rotation speed slowed because of the torque exerted on it by the magnetic field lines. In addition, much of the rotational angular momentum of the newly formed Sun was probably also carried away by the particles in the solar wind (you may recall a similar discussion in Chapter 11; see Fig. 11.27). In support of these mechanisms is Fig. 23.8. The change in slope of the angular-momentum-per-unit-mass curve corresponds well with the onset of surface convection in low-mass stars, which in turn is linked to the development of coronae and mass loss. Other mechanisms for angular momentum transport will be discussed later.

Composition Trends throughout the Solar System

We have already seen that lower-mass stars with metallicities similar to or greater than the solar value seem able to form planetary systems routinely. Therefore, the process of planetary system formation must be robust. The process must also be capable of producing systems with planets that are far from the parent star and systems where the planets are very close in.

A crucial piece of any successful theory must be the ability to explain the clear composition trends that exist among the planets in our Solar System (see, for example, Table 19.1). The inner terrestrial planets are small, generally volatile-poor, and dominated by rocky material, while the gas and ice giants contain an abundance of volatile material. Moreover, even though the ice giants Uranus and Neptune contain substantial volatiles, the gas giants Jupiter and Saturn contain the overwhelming majority of volatile material in the Solar System.

The moons of the giant planets also exhibit composition trends. In going from Jupiter out to Neptune, the progression is from rocky moons to increasingly icy bodies, first containing water-ice and then methane- and nitrogen-ice. The pattern even includes such objects as the asteroids, the Centaurs, the Kuiper belt objects, and other cometary nuclei. It is particularly important to note that a composition trend also exists across the asteroid belt itself. Even on the smaller scale of Jupiter's system of satellites, the Galilean moons change from volcanic Io to the thick-ice surface of Callisto.

The Temperature Gradient in the Solar Nebula

Apparently, either a composition gradient or a temperature gradient (or both) must have existed in the early solar nebula while these objects were forming. For instance, the observations just described could be accounted for if the temperature of the nebular disk had decreased sufficiently across the asteroid belt. In that case, water would not have condensed in the region of the terrestrial planets but could have condensed in the form of ice in the vicinity of the giant planets. Another temperature gradient associated with the formation of Jupiter could help to explain the formation of the Galilean moons from the Jupiter subnebula.

Recall from Section 18.2 that an accretion disk that forms in a binary star system has a well-determined temperature gradient [$T \propto r^{-3/4}$; see Eq. (18.17)]. An analogous sort

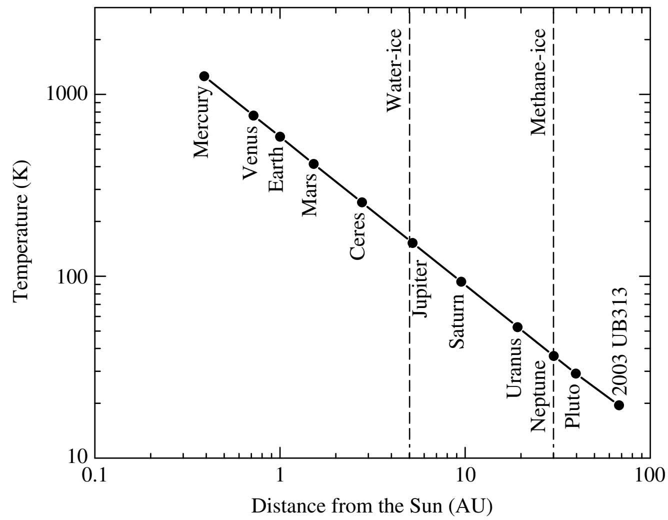

FIGURE 23.9 An equilibrium model of the temperature structure of the early solar nebula. Water-ice was able to condense out of the nebula in those regions beyond approximately 5 AU, and methane-ice could condense out of the nebula beyond 30 AU. The positions of the planets and Ceres represent their present-day locations.

of temperature structure should have existed in the solar nebula as well. The temperature structure for an equilibrium solar nebula model is shown in Fig. 23.9. Even though specific features of the distribution may change with more sophisticated modeling (by including time dependence, turbulence, and magnetic fields), it seems apparent that the condensation temperature of water-ice must be reached at some point near the current position of Jupiter, perhaps in the outer portions of the main asteroid belt (roughly 5 AU). The position in the solar nebula where water-ice could form has been variously referred to as the "snow line," the "ice line," or, more dramatically, the "blizzard line."

We have also learned that the environments around newly forming stars can be very dynamic places, with mass accretion and mass loss happening at virtually the same time in T-Tauri systems. During FU Orionis events, the environment around the star can become particularly active, with significant outbursts of energy occurring because of greatly increased mass accretion rates. It also seems certain that these environments will have complex magnetic fields that would lead to frequent and intense flares, analogous to the solar flares on our Sun that are produced by magnetic field reconnection events (Section 11.3).

Consequences of Heavy Bombardment

We have also seen throughout Part III that, at least within our own Solar System, the formation of the Sun was accompanied by the formation of a wide range of objects, including small rocky planets, gas giants, ice giants, moons, rings, asteroids, comets, Kuiper belt objects, meteoroids, and dust.

Of course, it is readily apparent that our Solar System is riddled with evidence of collisions in the past, leaving cratered surfaces on objects of all sizes, from planets and moons to asteroids and comets. As a consequence, any formation theory must also be able to account for the obvious, heavy bombardment endured by bodies in the early Solar System. As discussed in Chapter 20, the high mass density of Mercury and the extremely volatile-poor composition of the Moon strongly suggest that both of these worlds were directly influenced by cataclysmic collisions involving very large planetesimals (the Moon's formation is tied to just such a collision with Earth). Heavy surface cratering shows that collisions continued even after their surfaces formed, with a brief episode of late heavy bombardment about 700 Myr after the formation of the Moon. Features such as the enormous Herschel crater on Mimas and the bizarre surface of Miranda testify to the fact that the other bodies in the Solar System underwent the same intense barrage from planetesimals.

Another consequence of the heavy bombardment by planetesimals is the variety of present-day orientations for the spin axes of the planets. The extreme examples of the retrograde rotations of Uranus and Pluto have already been discussed, but the other planets must have had their rotation axes shifted as well. Assuming that the planets did form out of a flattened nebular disk, the inherent angular momentum of the system would have resulted in rotation axes being initially aligned nearly perpendicular to the plane of the disk. Because this is not the case today, some event (or events) must have occurred to alter the directions of the planets' rotational angular momentum vectors. With the exception of Venus's and Mars's complex tidal interactions with the Sun and the other planets, the only likely mechanism suggested to date that can naturally account for the range of orientations observed requires collisions of planets or protoplanets with large planetesimals.

The Distribution of Mass within Planetary Systems

Other features of the present-day Solar System that should be explained in a model of Solar System formation include the relatively small mass of Mars compared with its neighbors, the very small amount of mass present in the asteroid belt, and the existence of the Oort cloud and the Kuiper belt.

Furthermore, if we are to seek a general, unifying model of planetary system formation that includes our own Solar System as one example, it is necessary to understand the distributions of planets in other systems. Particularly perplexing when first discovered was the existence of "hot Jupiters" such as 51 Peg b. How could a gas giant form and survive so close to its parent star? In our own Solar System, none of the giant planets resides closer to the Sun than 5.2 AU.

Formation Timescales

One aspect of all formation theories that cannot be neglected are constraints imposed by timescales:

- The collapse of a molecular cloud was discussed in Section 12.2. In Fig. 12.9 we saw that once a collapse is initiated, on the order of 10^5 years is required for the formation of a protosun and nebular disk.

- The onset of violent T-Tauri and FU Orionis activity and extensive mass loss follows the initial collapse in some 10^5 to 10^7 years (see Section 12.3). This means that any nebular gas and dust that has not been accreted into a planetesimal or a protoplanet will be swept away within about 10 Myr, terminating further formation of large planets.

- The presence of $^{26}_{13}\mathrm{Al}$ in carbonaceous chondrites indicates that these meteorites must have been formed within a few million years after the creation of the aluminum, whether it was through a supernova detonation or through flares during FU Orionis activity. Otherwise, all of the radioactive nuclides that were created would have decayed into $^{26}_{12}\mathrm{Mg}$. This observation puts severe constraints on condensation rates in the early solar nebula.

- The oldest meteorites, including Allende, date back to near 4.566 Gyr, while the age of the Sun itself is 4.57 Gyr. Clearly these oldest meteorites must have formed rapidly within the solar nebula.

- The ages of rocks returned from the Moon show that the surface of that body must have solidified some 100 Myr after the collapse of the solar nebula. Similar constraints exist on the formation of the surface of Mars judging on the basis of the age of the Martian meteorite, ALH84001.

- The lunar surface underwent a spike of late heavy bombardment about 700 Myr after the Moon formed.

- As we will learn later, as planets grow in accretion nebulae, they tend to migrate inward due to tidal interactions with the nebula and viscosity effects. It is estimated that a planetesimal could drift all the way into its parent star from a distance of 5 AU within roughly 1 to 10 Myr.

- A rather loose constraint on any model requires that all of the planets, moons, asteroids, Kuiper belt objects, and comets must be fully formed today, 4.57 Gyr years after the process started. Although this may seem trivial, not all models of Solar System formation have been successful in creating planets this rapidly!

The Gravitational Instability Formation Mechanism

Two general, competing mechanisms have been proposed for the formation of planets within the accretion disks of proto- and pre-main-sequence stars. One mechanism is based on the idea that planets (or perhaps brown dwarfs) could form in accretion disks in a manner analogous to star formation. In regions where there may be a greater density of material in the disk, self-collapse could result. As the mass accumulates in that region, its gravitational influence on the surrounding disk increases, causing additional material to accrete onto the newly forming planet. This mechanism could even result in a local subnebula accretion disk forming around the protoplanet that could lead to the creation of moons and/or ring systems.

While this "top-down" **gravitational instability** mechanism has several attractive features, including simplicity and being strongly analogous to the formation of protostars, its general applicability suffers from numerous difficulties. By observations of other accretion disks, along with T-Tauri accretion and mass-loss rates, and combined with detailed numerical simulations, it appears that the solar nebula's lifetime would not have been sufficient to allow objects like Uranus and Neptune to grow quickly enough to attain the masses we observe before the nebula was depleted. This mechanism also does not explain the large number of other, smaller objects that are present in our Solar System and are likely to exist in other planetary systems as well (recall the β Pic debris disk). In addition, the gravitational instability mechanism doesn't appear to readily account for the mass distribution of extrasolar planets, the correlation between planetary system formation and metallicity, or the wide range in the densities and core sizes of planets, both within our Solar System and among the extrasolar planets.

The Accretion Formation Mechanism

An alternative model, and the one general favored by most astronomers, is that planets grow from the "bottom up" through a process of **accretion** of smaller building blocks. Based on all of the observational and theoretical information presently available, it appears that a reasonable description of the formation of planetary systems can now be given. What follows is a possible scenario for the formation of our own Solar System, although references to general aspects of planetary system formation will also be made. It is important to note, however, that because of the complexity of the problem, revisions in the model (both minor and major) are likely to occur in the future.

The Formation of the Solar System: An Example

Within an interstellar gas and dust cloud (perhaps a giant molecular cloud), the Jeans condition was satisfied locally, and a portion of the cloud began to collapse and fragment (see Eq. 12.14 for the Jeans mass). The most massive segments evolved rapidly into stars on the upper end of the main sequence, while less massive pieces either were still in the process of collapsing or had not yet started to collapse. Within a period of a few million years or less, the most massive stars would have lived out their entire lives and died in spectacular supernovae explosions.[2]

As the expanding nebulae from one or more of the supernovae traveled out through space at a velocity of roughly $0.1c$, the gases cooled and became less dense. It may have been during this time that the most refractory elements began to condense out of the supernova remnants, including calcium, aluminum, and titanium, the ingredients of the CAIs that would eventually be discovered in carbonaceous chondrites that would fall to Earth billions of years later. When a supernova remnant encountered one of the cooler, denser components of the cloud that had not yet collapsed, the remnant began to break up into "fingers" of gas and dust that penetrated the nebula unevenly. The small cloud fragment would have also been compressed by the shock wave of the high-speed supernova remnant when the expanding nebula collided with the cooler gas. It is possible that this compression may have even helped trigger the collapse of the small cloud. In any case, the material in the solar nebula was now enriched with elements synthesized in the exploded star.

Assuming that the solar nebula possessed some initial angular momentum, conservation of angular momentum demands that the cloud "spun up" as it collapsed, producing a protosun surrounded by a disk of gas and dust. In fact, the disk itself probably formed more rapidly than the star did, causing much of the mass of the growing protosun to be funneled through the disk first. Although this important point is not entirely resolved, it has

[2]Recall the extremely disparate timescales for high-mass and low-mass stellar evolution; see Tables 12.1 and 13.1.

been estimated that the solar nebular disk may have contained a few hundredths of a solar mass of material, with the remaining 1 $M_\odot$ of the nebula ending up in the protosun. At the very least, a minimum amount of mass must have ended up in the nebular disk to form the planets and other objects that exist today. Such a disk is referred to as the **Minimum Mass Solar Nebula** (see Problem 23.4).

The Hill Radius

Within the nebular disk, small grains with icy mantles were able to collide and stick together randomly. When objects of appreciable size were able to develop in the disk, they began to gravitationally influence other material in their areas.

To quantify the influence that these growing planetesimals had, we can define the **Hill radius**, R_H, to be that distance from the planetesimal where the orbital period of a test particle around the planetesimal is equal to the orbital period of the planetesimal around the Sun.

Assuming a circular orbit, the orbital period of a test particle (m_t) around an object of mass M ($M \gg m_t$) at a distance R is given by Kepler's third law (Eq. 2.37) as

$$P \simeq 2\pi\sqrt{\frac{R^3}{GM}}.$$

At a distance a from the Sun, the orbital period of the growing planetesimal around the Sun equals the orbital period of a massless test particle around the planetesimal at the Hill radius when

$$\sqrt{\frac{a^3}{M_\odot}} = \sqrt{\frac{R_H^3}{M}}.$$

Thus, the Hill radius is given by

$$\boxed{R_H = \left(\frac{M}{M_\odot}\right)^{1/3} a.} \tag{23.4}$$

Rewriting in terms of the density of the Sun and the density of the planetesimal (assumed to be spherical), the Hill radius becomes

$$R_H = R/\alpha \tag{23.5}$$

where R is the radius of planetesimal and

$$\alpha \equiv \left(\frac{\rho_\odot}{\rho}\right)^{1/3} \frac{R_\odot}{a}.$$

The physical significance of the Hill radius is that if a particle comes within about one Hill radius of a planetesimal with a relative velocity that is sufficiently low, the particle can become gravitationally bound to the planetesimal. In this way, the planetesimal acquires the mass of the particle and continues to grow. Of course, as the planetesimal's radius grows, so does its Hill radius.

Example 23.2.1. For a planetesimal of density $\rho = 800\,\mathrm{kg\ m^{-3}}$ and radius 10 km, located 5 AU from the Sun ($\rho_\odot = 1410\ \mathrm{kg\ m^{-3}}$), the planetesimal's Hill radius would be

$$R_H = R/\alpha = R\left(\frac{\rho}{\rho_\odot}\right)^{1/3}\left(\frac{a}{R_\odot}\right) = 8.9\times10^6\ \mathrm{m} = 1.4\ \mathrm{R}_\oplus.$$

This planetesimal is similar to present-day cometary nuclei.

The Formation of the Gas and Ice Giants

As the low-energy collisions continued, progressively larger planetesimals were able to form. In the innermost regions of the disk the accreting particles were composed of CAIs, silicates (some in the form of chondrules), iron, and nickel; relatively volatile materials were unable to condense out of the nebula because of the high temperatures in that region. At distances greater than 5 AU from the growing protosun, just inside the present-day orbit of Jupiter, the nebula became sufficiently cool that water-ice could form as well. The result was that water-ice could also be included in the growing planetesimals beyond that distance. Even farther out (perhaps near 30 AU, the present-day orbit of Neptune), methane-ice also participated in the development of planetesimals. The location of the "snow line" where water-ice could form is shown in Fig. 23.10 (recall also Fig. 23.9).

The object that grew most rapidly was Jupiter. Thanks to the presence of water-ice along with rocky materials, and with a nebula that was sufficiently dense in its region, Jupiter's core reached a mass of between 10 and 15 $\mathrm{M}_\oplus$. At that point the planet's gravitational influence became great enough that it started to collect the gases in its vicinity (principally hydrogen and helium). In effect, this created a localized subnebula, complete with its own accretion disk. The outcome was the formation of the massive planet we see today, together with the Galilean satellites. Heat generated in the gravitational collapse of Jupiter, combined with tidal effects, led to the eventual evolution of its moons. Astronomers believe that the entire process of forming Jupiter required on the order of 10^6 years, halting when the gas was depleted.

As we will see shortly, the formation of the massive Jupiter had a significant impact on the other three planetesimals that had also grown to significant size beyond the snow

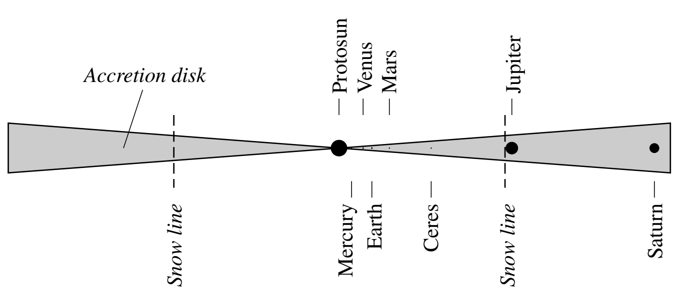

FIGURE 23.10 A schematic drawing of the solar nebular disk, indicating the position of the water-ice "snow line" 5 AU from the protosun. Methane-ice began forming at roughly 30 AU from the protosun as well. The protosun, the protoplanets, and Ceres are located at their relative present-day distances, but their relative sizes are not correct.

line. Although Saturn, Uranus, and Neptune all developed cores of 10 to 15 $M_\oplus$, they were somewhat farther out in the nebula where the density was lower. As a result, they were unable to acquire the amount of gas that Jupiter captured in the same period of time.

The Formation of the Terrestrial Planets and the Asteroids

In the inner portion of the solar nebula the temperatures were too warm to allow the volatiles to condense out and participate in the formation of planetesimals. But as the nebula cooled, the most refractory elements were able to condense out to form the CAIs.[3] Next to condense were the silicates and other equally refractory materials.

The slow relative velocities of silicate grains in nearly identical orbits resulted in low-energy collisions that promoted grain growth. Eventually, a hierarchy of planetesimal sizes developed. Computer simulations suggest that in the region of the terrestrial planets, along with a large number of smaller objects, there may have been as many as 100 planetesimals roughly the size of the Moon, 10 with masses comparable to Mercury's, and several as large as Mars. However, during the accretion process, most of these large planetesimals became incorporated into Venus and Earth. When the forming planets became massive enough, internal heat that was generated by decaying radioactive isotopes, together with energy released during collisions, started the process of gravitational separation. The results were the chemically differentiated worlds we see today.

With the formation of the massive Jupiter just beyond 5 AU from the Sun, gravitational perturbations began to influence the orbits of planetesimals in the region. In particular, most of the objects in the present-day asteroid belt had their orbits "pumped up" into progressively more and more eccentric orbits until some of them were absorbed by Jupiter or the other developing planets or were sent crashing into the Sun, while most were ejected from the Solar System entirely. This process stole material from the "feeding zones" near Mars and in the asteroid belt, resulting in a smallish fourth planet and very little mass in the belt. Perhaps only 3% of the original mass near Mars's orbit remained and only 0.02% of the mass in the region of the belt. Continued perturbations from Jupiter meant that the remaining belt of planetesimals had rather high relative velocities and were never able to consolidate into a single object. In fact, the high relative velocities imply that collisions cause fracturing, rather than growth.

As planetesimals continued to move throughout the forming Solar System, other collisions occurred. Some of the largest planetesimals in the inner Solar System collided with Mercury, removing its low-density mantle, and some struck Earth, forming the Earth–Moon system. Still other planetesimals of significant mass crashed into Mars and the outer planets, changing the orientations of their axes. Apparently, some of the planetesimals were also captured as moons or were torn apart by the giant planets when they wandered inside the planets' Roche limits.

Long before the terrestrial planets finished "feeding" on planetesimals in their regions of the disk, however, the evolving Sun reached the stage of thermonuclear ignition in its core, initiating the T-Tauri phase. At this point the infall of material from the disk was reversed by the strong stellar wind that ensued, and any gases and dust that had not yet collected into planetesimals were driven out of the inner Solar System.

[3]It is only in the innermost part of the Solar System that the nebula was warm enough to form CAIs in the first place.

The Process of Migration

The accretion scenario described above is not without its own challenges. For instance, a long-standing problem has to do with the formation of the ice giants. At their current positions in the Solar System, it appears that the solar nebula would not have been dense enough to allow them to reach their present-day masses before the remaining gas was swept away by the T-Tauri wind. In addition, how is the episode of late heavy bombardment to be explained as a spike in collision rates roughly 3.8 Gyr ago? The apparent solution to both of these problems seems to lie in understanding a perplexing problem with many extrasolar planets.

With the discovery of “hot Jupiters” in extrasolar planetary systems, scientists realized that planets must be able to migrate inward while they are forming, and Jupiter is no different. Computer simulations of Solar System evolution suggest that Jupiter formed about 0.5 AU farther out in the nebula than its current position.

One mechanism by which inward migration of Jupiter (and extrasolar planets) could occur involves gravitational torques between the planet and the disk.[4] In this mechanism, initial deviations from axial symmetry produce density waves in the disk (density waves were mentioned in Section 21.3 in connection with the dynamics of Saturn’s rings and will be discussed again extensively in connection with galaxy dynamics; see Section 25.3). The gravitational interaction between a growing planet and density waves results in the simultaneous transfer of angular momentum outward and mass inward. This so-called **Type I migration** mechanism can be shown to be proportional to mass, implying that as the planet accretes more material, it moves more rapidly toward its parent star. It may be that this can actually cause some planets to collide with the star on a timescale of one to ten million years.

However, it initially appeared that the timescale for Type I migration was too short compared with the runaway accretion of gases onto the growing Jupiter; in other words, Jupiter would crash into the Sun before it could fully form. It also appeared that Jupiter couldn’t grow rapidly enough to reach its present size before the nebula was dissipated by the T-Tauri wind.

The solution to these problems may rest with the migration process itself. As the growing planet moves through the solar nebula, it continually encounters fresh material to “feed on.” If the planet remained in a fixed orbit, it would quickly consume all of the available gas within several Hill radii and would grow only slowly after that. Migration allows it to move through the disk without creating a significant gap in the nebula.

It has also been shown that viscosity within the disk can cause objects to migrate inward. This **Type II migration** mechanism causes slowly orbiting particles farther out to speed up because of collisions with higher-velocity particles occupying slightly smaller orbits.[5] The loss of kinetic energy by the inner particles causes them to spiral inward. Type II migration can become the more significant, if slower, migration process when a gap is opened up in the disk.

[4]Peter Goldreich and Scott Tremaine suggested in 1980 that this mechanism would be important in the dynamical evolution of accretion disks. Their paper was published some fifteen years before the first confirmed detection of an extrasolar planet.

[5]Recall that the problem of angular momentum transport in a disk was also discussed in Chapter 18.

Outward migration is also possible. In this case, the scattering of planetesimals inward results in migration outward. Whether inward or outward migration occurs depends on the density of the nebula and the abundance of planetesimals.

Applying the mechanisms of migration to the evolution of our own Solar System, it appears that Jupiter not only influenced objects interior to its present-day orbit but also was influential in causing Saturn, Uranus, and Neptune to migrate outward. It seems that Uranus and Neptune initially formed their cores in a region of the nebula with a greater density, just as Jupiter and Saturn did. However, because of outward migration, they were able to put on only a small amount of extra gas and remain today as ice giants, rather than gas giants.

Resonance Effects in the Early Solar System

Assuming that Jupiter originally formed at about 5.7 AU from the Sun as some simulations suggest, and that Saturn formed perhaps 1 AU closer to the Sun than its current position, the two gas giants would have moved through a critical resonance as Jupiter migrated inward and Saturn migrated outward. When the orbital periods of the two planets reached a 2:1 resonance (i.e., the orbital period of Saturn was exactly twice the orbital period of Jupiter), their gravitational influences on other objects in the Solar System would have periodically combined at the same points in their orbits, causing significant perturbations to orbits of objects in the asteroid belt and in the Kuiper belt.[6] Computer simulations suggest that this resonance effect may have occurred about 700 Myr after the formation of the inner planets and our Moon. It seems plausible that the passage of Jupiter and Saturn through this 2:1 resonance may have caused the episode of late heavy bombardment that is now recorded on the surface of the Moon.

As a consequence of Neptune's outward migration, Neptune swept up some of the remaining planetesimals, trapping them in 3-to-2 orbital resonances with the planet as it moved outward. It may be that Pluto and the other Plutinos were caught up in this outward migration. The orbits of the scattered Kuiper belt objects were also likely to have been perturbed by the migration of Neptune. The classical KBOs were probably far enough from Neptune not to be as drastically affected by its migration. In fact, the Kuiper belt may be the Solar System's analog to debris disks seen around other stars.

Similarly, the Oort cloud cometary nuclei are likely to be planetesimals that were scattered more severely by Uranus and Neptune. Once sufficiently far from the Sun, scattered cometary nuclei had their orbits randomized by passing stars and interstellar clouds.

The Formation of CAIs and Chondrules

A particularly challenging problem with the model of Solar System formation described above is the presence of chondrules mixed in with CAIs in a matrix of hydrated and carbon-bearing minerals in chondritic meteorites. Both the chondrules and the CAIs have certainly been exposed to intense heat, but the matrix has clearly never been heated to temperatures greater than a few hundred kevins. Because silicates require lower temperatures to condense out of the solar nebula, the chondrules probably formed after the CAIs. Silicate dust grains likely formed out of the nebula, coalescing into small clumps through repeated collisions.

[6]Recall the effect of Mimas on the ring system of Saturn (the Cassini division) and the effect of Jupiter on the asteroid belt (the Kirkwood gaps).

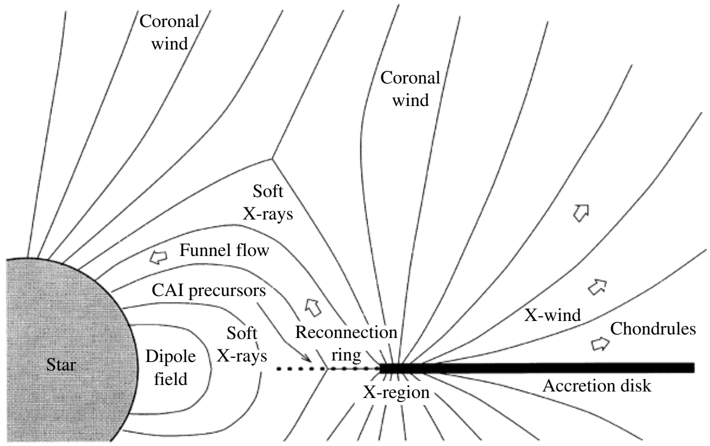

FIGURE 23.11 A schematic diagram of the X-wind model. (Figure adapted from Shu et al., *Ap. J.*, *548*, 1029, 2001.)

However, they could not have formed initially as molten droplets but, instead, were melted after formation.

Currently, the most plausible scenario suggests that powerful flares during FU Orionis events may be responsible for the melting or partial melting of chondrules and CAIs. As the inner edge of the accretion disk moves in and out on timescales of 30 years or so (perhaps associated with magnetic field activity), the silicate grains are exposed to flash heating by flares resulting from reconnection events. In the rarefied environment at the interior edge of the nebula, the droplets are able to cool rather quickly, perhaps between 100 and 2000 K per hour. Frank Shu and his colleagues have suggested that the metamorphosed chondrules may be launched back into the planet-forming region of the nebula by an X-wind, similar to the wind responsible for the ejection of Herbig–Haro objects in jets (see Fig. 23.11). In the planet-forming region of the nebula, the chondrules and CAIs are incorporated into the matrix. This model of chondrule formation implies that the solar nebula was a very dynamic system indeed.

Although much work remains to be done in fully developing our understanding of Solar System formation and the formation extrasolar planetary systems, tremendous progress has been made in this very complex area of research.

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PROBLEMS

23.1 **(a)** If the actual separation between HD 80606 and HD 80607 is 2000 AU, determine the orbital period of the binary star system. *Hint:* You may want to refer to the data in Appendix G to estimate the masses of the two stars.

(b) How many orbits will the planet HD 80806b make around its parent star in the time that the two stars complete one orbit about their common center of mass? The semimajor axis of the planet's orbit is 0.44 AU.

(c) What is the ratio of the force of gravity exerted on HD 80606b by its parent star to that exerted by HD 80607 when it is aligned between the two stars?

23.2 Compare Pluto with asteroids and cometary nuclei. Comment on the significance of any differences in view of the evolutionary model discussed in the chapter.

23.3 **(a)** If all of the angular momentum that is tied up in the rest of the Solar System could be returned to the Sun, what would its rotation period be (assume rigid-body rotation)? Refer to the data in Fig. 23.8. The moment-of-inertia ratio of the Sun is 0.073.

(b) What would the equatorial velocity of the photosphere be?

(c) How short could the rotation period be before material would be thrown off from the Sun's equator?

23.4 The Minimum Mass Solar Nebula is the smallest nebula that could be formed and still have sufficient mass to create all of the objects in the Solar System. Make a rough estimate of the mass of the Minimum Mass Solar Nebula.

23.5 Estimate the present-day Hill radius of Jupiter. Express your answer in terms of the radius of Jupiter, as well as in astronomical units.

23.6 HD 63454 is a K4V star known to have an extrasolar planet orbiting it in a circular orbit with an orbital period of 2.81782 d, producing a maximum radial reflex velocity of 64.3 m s^{-1} relative to the center of mass of the star. The distance to HD 63454 is 35.80 pc. Consulting Appendix G, determine

(a) the semimajor axis of the planet's orbit.

(b) the minimum mass of the planet.

(c) the maximum astrometric wobble of the star due to the planet's pull, expressed in arcseconds.

23.7 14 Her is a K0V star located 18.1 pc from Earth. The extrasolar planet orbiting the star has an orbital period of 1796.4 d with an orbital eccentricity of 0.338. Consulting Appendix G, determine

(a) the semimajor axis of the planet's orbit.

(b) the maximum separation of the planet from the center of mass of its parent star.

(c) the velocity of the planet in its orbit at closest approach to the star.

23.8 Explain why the high metallicities of systems with known extrasolar planets support the hypothesis that planets form from the "bottom up" by mass accretion of planetesimals.

23.9 Assume that Jupiter and Saturn formed 5.7 AU and 8.6 AU from the Sun, respectively. Show that if the planets simultaneously migrated to their present orbital distances, they passed through a 2:1 orbital period resonance.

PART IV

Galaxies and the Universe

CHAPTER 24

The Milky Way Galaxy

24.1 *Counting the Stars in the Sky*
24.2 *The Morphology of the Galaxy*
24.3 *The Kinematics of the Milky Way*
24.4 *The Galactic Center*

24.1 ■ COUNTING THE STARS IN THE SKY

As we learned in the first two chapters of this text, human beings have long looked up at the heavens and contemplated its vastness, proposing various models to explain its form. In some civilizations the stars were believed to be located on a celestial sphere that rotated majestically above a fixed, central Earth. When Galileo made his first telescopic observations of the night sky in 1610, we started down a long road that has dramatically expanded our view of the universe.

In this chapter we will explore the complex system of stars, dust, gas, and dark matter known as the **Milky Way Galaxy**.[1] Although it is possible to get at least a general idea about the nature of other galaxies from our external viewpoint, studying our own Galaxy has proved to be very challenging. As we will learn, we live in a disk of stars, dust, and gas that severely impacts our ability to "see" beyond our relative stellar neighborhood when we look along the plane of the disk. The problem is most severe when looking toward the center of the Galaxy in the constellation Sagittarius. In Section 24.1 we will discover that studying the distribution of stars while considering the effects of extinction provides us with our first hint of what the Milky Way looks like from an outside perspective. In Section 24.2, a detailed description of the many varying components of the Galaxy will be presented.

Much of what we know today about the formation and evolution of the Milky Way is encoded in the motions of the Galaxy's constituents, especially when combined with information about variations in composition. Unfortunately, measuring the motions of the stars and gas in the Galaxy is done from an observing platform (Earth) that is itself undergoing a complex motion that involves the orbit of Earth around the Sun and the Sun's elaborate path around the Galaxy. In Section 24.3 we will investigate these motions, allowing us to move from a description of motions relative to the Sun to motions relative to the center of

[1]Throughout the remainder of this text, we may refer to the Milky Way Galaxy alternately in the shortened forms "the Galaxy" and "our Galaxy."

the Galaxy. We will also be led to the remarkable conclusion that the luminous, baryonic matter in the Galaxy is only a small fraction of what our Galaxy is composed of.

Finally, in Section 24.4, we will probe the center of the Milky Way and study the exotic environment found there, including indisputable evidence for a supermassive black hole.

Throughout the remainder of Part IV we will investigate other, more distant galaxies; we will study their morphologies and evolution, as well as the evolution of our own Milky Way. We will also study the large-scale structure of the universe and trace our developing understanding of the earliest moments of the universe and its ultimate fate.

Historical Models of the Milky Way Galaxy

As can be seen by even a casual observation of the dark night sky, an almost continuous band of light appears to circle Earth, inclined about 60° with respect to the celestial equator (see Fig. 24.1). It was Galileo who first realized that this Milky Way is a vast collection of individual stars. In the mid-1700s, in order to explain its circular distribution across the heavens, Immanuel Kant (1724–1804) and Thomas Wright (1711–1786) proposed that the Galaxy must be a stellar disk and that our Solar System is merely one component within that disk. Then, in the 1780s, William Herschel (1738–1822) produced a map of the Milky Way based crudely on counting the numbers of stars that he could observe in 683 regions of the sky (see Fig. 24.2). In his analysis of the data, Herschel assumed that (a) all stars

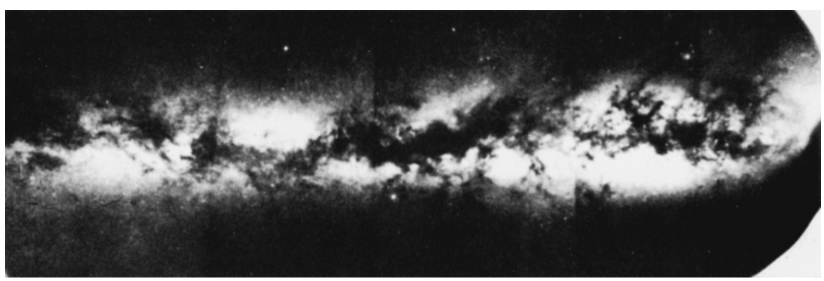

FIGURE 24.1 A mosaic of the Milky Way showing the presence of dust lanes. (Courtesy of The Observatories of the Carnegie Institution of Washington.)

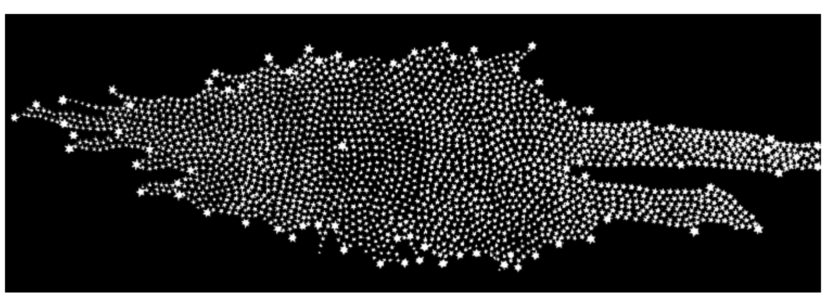

FIGURE 24.2 William Herschel's map of the Milky Way Galaxy, based on a qualitative analysis of star counts. He believed that the Sun (indicated by a larger star) resided near the center of the stellar system. (Courtesy of Yerkes Observatory.)

have approximately the same absolute magnitude, (b) the number density of stars in space is roughly constant, (c) there is nothing between the stars to obscure them, and (d) he could see to the edges of the stellar distribution. From his data, Herschel concluded that the Sun had to be very near the center of the distribution and that the dimensions measured along the plane of the disk were some five times greater than the disk's vertical thickness.

Jacobus C. Kapteyn (1851–1922) essentially confirmed Herschel's model of the Galaxy, again using the technique of star counting. However, through the use of more quantitative methods, Kapteyn was able to specify a distance scale for his model of the Galaxy. The **Kapteyn universe**, as it is now called, was a flattened spheroidal system with a steadily decreasing stellar density with increasing distance from the center. A depiction of the Kapteyn universe is shown in Fig. 24.3. In the plane of the Galaxy and at a distance of some 800 pc from the center, the number density of stars had decreased from its central value by a factor of two. On an axis passing through the center and perpendicular to the central plane, the number density decreased by 50% over a distance of only 150 pc. The number density diminished to 1% of its central value at distances of 8500 pc and 1700 pc in the plane and perpendicular to the plane, respectively. Kapteyn concluded that the Sun was located 38 pc north of the Galactic midplane and 650 pc from the center, measured along the Galactic midplane.

To get some idea of how Kapteyn arrived at his nearly heliocentric (or Sun-centered) model of the universe, recall the equation for the distance to a star whose absolute magnitude is known (Eq. 3.5):

$$d = 10^{(m-M+5)/5}.$$

Assuming a value for M (for instance, if the spectral class and luminosity class are known), and measuring m at a telescope, the distance modulus $m - M$ and the distance d are readily obtained. And given the known coordinates of the star on the celestial sphere, its three-dimensional position relative to Earth is determined.

In actuality, since the number of stars in any given region is so great, it is impractical to estimate the distance to each individual star in the way just described. Instead, a statistical approach is used that is based on counting the number of stars in a specified region down to a predetermined limiting apparent magnitude. From this counting procedure, the number density of stars at a given distance from the Sun can be estimated. We will discuss some

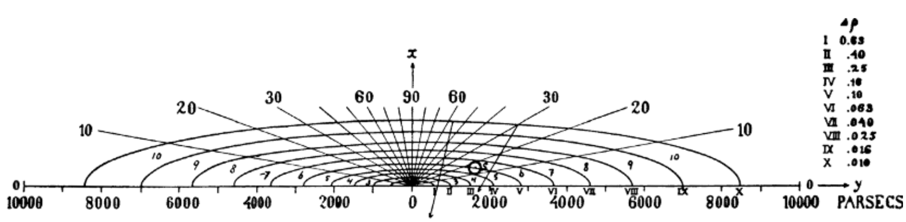

FIGURE 24.3 The Kapteyn universe. Surfaces of constant stellar number density are indicated around the Galactic center. Note that the open circle does not represent the position of the Sun derived by Kapteyn. Rather, the open circle was used as an estimate from which Kapteyn began his analysis of the available data. (Figure from Kapteyn, *Ap. J.*, *55*, 302, 1922.)

of the details of this method shortly. In his original study, Kapteyn used over 200 selected regions of the sky.

During the years between 1915 and 1919, shortly before Kapteyn's model was published, Harlow Shapley (1885–1972) estimated the distances to 93 globular clusters using RR Lyrae and W Virginis variable stars (Section 14.1). Since these stars are easily identified in the clusters through their periodic variations in luminosity, it is a relatively simple matter to use their absolute magnitudes (obtained from a period–luminosity relation such as Eq. 14.1) to estimate their distances from the Sun. The distances to the variable stars correspond to the distances to the clusters in which they reside.

In analyzing his data, Shapley recognized that the globular clusters are not distributed uniformly throughout space but are found preferentially in a region of the sky that is centered in the constellation of Sagittarius, at a distance that he determined to be 15 kpc from the Sun. Furthermore, he estimated that the most distant clusters are more than 70 kpc from the Sun, over 55 kpc beyond the center. As a result, by assuming that the extent of the globular clusters was the same as the rest of the Galaxy, Shapley believed that the diameter of the Galaxy was on the order of 100 kpc, close to ten times the value proposed by Kapteyn. Shapley's picture of the Galaxy also differed from Kapteyn's in another important way: Kapteyn's model located the Sun relatively near the center of the distribution of stars, whereas Shapley's Galactic center was much farther away.

We know today that both Kapteyn and Shapley were in error; Kapteyn's universe was too small and the Sun was too near the center, and Shapley's Galactic model was too large. Surprisingly, both models erred in part for the same reason: the failure to include in their distance estimates the effects of interstellar extinction due to dust and gas. Kapteyn's selected regions were largely within the Galactic disk where extinction effects are most severe; as a result, he was unable to see the most distant portions of the Milky Way, causing him to underestimate its size. The problem is analogous to someone on Earth trying to see the surrounding land while standing in a dense fog with limited visibility. Shapley, on the other hand, chose to study objects that are generally found well above and below the plane of the Milky Way and that are inherently bright, making them visible from great distances. It is in directions perpendicular to the disk that interstellar extinction is least important, although it cannot be neglected entirely. Unfortunately, errors in the calibration of the period–luminosity relation used by Shapley led to overestimates of the distances to the clusters. The calibration errors were traced to the effects of interstellar extinction (see Section 27.1).

Interestingly, Kapteyn was aware of the errors that interstellar extinction could introduce, but he was unable to find any quantitative evidence for the effect, even though researchers suspected that dust might be responsible for the dark bands seen running across the Milky Way (see Fig. 24.1). Further evidence for strong extinction could also be found in Shapley's own data; no globular clusters were visible within a region between approximately $\pm 10^\circ$ of the Galactic plane, called the **zone of avoidance**. Shapley suggested that globular clusters were apparently absent in the zone of avoidance because strong gravitational tidal forces disrupted the objects in that region. In reality, interstellar extinction is so severe within the zone of avoidance that the very bright clusters are simply undetectable. Clearly the problems encountered by Kapteyn and Shapley in deducing the structure of our Galaxy point out the difficulty of determining its general morphology from a nearly fixed location within the

Galaxy's disk. Unfortunately, given the immense distances involved, we are not likely to be able to reach another, more favorable vantage point any time soon!

The Effects of Interstellar Extinction

To see how interstellar extinction directly affects estimates of stellar distances, we must modify Eq. (3.5), as was done in Section 12.1. Starting from Eq. (12.1) and solving for d, we find

$$d = 10^{(m_\lambda - M_\lambda - A_\lambda + 5)/5} = d' 10^{-A_\lambda/5}, \tag{24.1}$$

where $d' = 10^{(m_\lambda - M_\lambda + 5)/5}$ is the erroneous estimate of distance made when extinction is neglected, and A_λ is the amount of extinction in magnitudes, and as a function of wavelength, that has occurred between the star and Earth. Since $A_\lambda \geq 0$ in all cases (after all, extinction cannot make a star appear brighter), $d \leq d'$; the true distance is always less than the apparent distance.

In the disk of the Milky Way the typical rate of extinction in visible wavelengths is 1 magnitude kpc^{-1}, although that value can vary dramatically if the line of sight includes distinct nebulae such as giant molecular clouds. Fortunately, it is often possible to estimate the amount of extinction by considering how dust affects the color of a star (interstellar reddening); see Section 12.1.

Example 24.1.1. Suppose that a B0 main-sequence star with an absolute visual magnitude of $M_V = -4.0$ is observed to have an apparent visual magnitude of $V = +8.2$. Neglecting interstellar extinction (i.e., assuming naively that $A_V = 0$), the distance to the star would be estimated to be

$$d' = 10^{(V - M_V + 5)/5} = 2800 \text{ pc}.$$

However, if it is known by some independent means (such as reddening) that the amount of extinction along the line of sight is 1 mag kpc^{-1}, then $A_V = kd$ mag, where $k = 10^{-3}$ mag pc^{-1} and d is measured in pc. This gives

$$d = 10^{(V - M_V - kd + 5)/5} = 2800 \times 10^{-kd/5} \text{ pc},$$

which may be solved iteratively or graphically, giving a true distance to the star of $d = 1400$ pc.

In this case the distance to the star would have been overestimated by almost a factor of two if the effects of interstellar extinction were not properly accounted for.

Differential and Integrated Star Counts

As we have already mentioned, Kapteyn's method of star counting was not based on directly determining d for individual stars. Rather, the numbers of stars visible in selected regions of the sky are counted over a specified apparent magnitude range. Alternatively, all stars in the regions brighter than a chosen limit of apparent magnitude can be counted. These approaches are known as **differential** and **integrated star counts**, respectively. The technique of star counting is still used today to determine the number density of stars in the sky. The distribution depends on a variety of parameters, including direction, distance, chemical composition, and spectral classification. Such information is very helpful to astronomers in their efforts to understand the structure and evolution of the Milky Way Galaxy.

Let $n_M(M, S, \Omega, r)\, dM$ be the number density of stars with absolute magnitudes between M and $M + dM$ and attribute S that lie within a solid angle Ω in a specific direction at a distance r from the observer (S could be composition or the Morgan–Keenan spectral class discussed in Section 8.2, for example). According to the notation used here, n_M has units of pc^{-3}mag^{-1}, and the actual number density of stars having attribute S that lie within a solid angle Ω and located a distance r from the observer is given by

$$n(S, \Omega, r) = \int_{-\infty}^{\infty} n_M(M, S, \Omega, r)\, dM. \tag{24.2}$$

If other specific information is also desired, those attributes could be included in S as well. However, the amount of data required to carry out star count analyses is formidable and often prohibitive; as the number of variables increases, so does the amount of data required. In his original study, Kapteyn considered *general* star counts that tracked absolute magnitude only, regardless of spectral class.

If the number density $n_M(M, S, \Omega, r)\, dM$ is integrated over the volume of a cone defined by the solid angle Ω and extending from the observer at $r = 0$ to some distance $r = d$, the result is $N_M(M, S, \Omega, d)\, dM$, the total number of stars with absolute magnitudes in the range M to $M + dM$ that are found within that conical volume of space (see Fig. 24.4). Using $dV = \Omega r^2\, dr$ in spherical coordinates, this is

$$N_M(M, S, \Omega, d)\, dM = \left[\int_0^d n_M(M, S, \Omega, r)\, \Omega r^2\, dr\right] dM. \tag{24.3}$$

Equation (24.3) is the general expression for the *integrated star count*, written in terms of

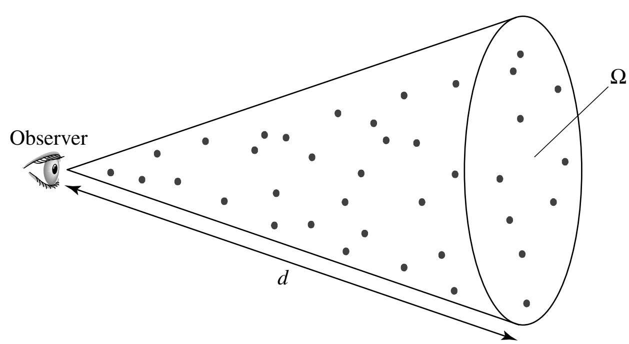

FIGURE 24.4 An observer on Earth counting the number of stars within a specified range of Morgan–Keenan spectral and luminosity types out to a distance d located within a cone of solid angle Ω. When the effects of interstellar extinction are considered, this is equivalent to counting the same group of stars down to a corresponding apparent magnitude, m.

the limiting distance, d. Note that this means that $n_M\,dM$ can be obtained from $N_M\,dM$ (with limiting distance r) by differentiating:

$$n_M(M, S, \Omega, r)\,dM = \frac{1}{\Omega r^2}\frac{dN_M\,dM}{dr}.$$

Of course, those stars sharing the same absolute magnitude will have different apparent magnitudes because they lie at different distances from us. We can use Eq. (24.1) to replace the limiting distance, d, with the apparent magnitude m. This results in $\overline{N}_M(M, S, \Omega, m)\,dM$, the integrated star count written in terms of the limiting magnitude, m. (Note that the "bar" designation now indicates that the integrated star count is a function of m rather than d.) Thus $\overline{N}_M(M, S, \Omega, m)\,dM$ is the total number of stars with absolute magnitudes in the range M to $M + dM$ that appear brighter than the limiting magnitude, m.

If the limiting magnitude is increased slightly, then the limiting distance becomes correspondingly greater and the conical volume of space is extended to include more stars. The increase in the number of included stars is

$$\left[\frac{d\overline{N}_M(M, S, \Omega, m)}{dm}\,dm\right]dM.$$

This defines the *differential star count*,

$$A_M(M, S, \Omega, m)\,dM\,dm \equiv \frac{d\overline{N}_M(M, S, \Omega, m)}{dm}\,dM\,dm, \tag{24.4}$$

the number of stars with an absolute magnitude between M and $M + dM$ that are found within a solid angle Ω and have apparent magnitudes in the range between m and $m + dm$.

As a simple (and unrealistic) illustration of the use of integrated and differential star counts, consider the case of an infinite universe of uniform stellar density [i.e., $n_M(M, S, \Omega, r) = n_M(M, S) = \text{constant}$] and no interstellar extinction ($A = 0$). Then Eq. (24.3) becomes, after canceling dM,

$$N_M(M, S, \Omega, d) = n_M(M, S)\,\Omega\int_0^d r^2\,dr = \frac{\Omega\,d^3}{3}n_M(M, S).$$

(Note that when the last expression is considered over all directions, $\Omega = 4\pi$ and $\Omega d^3/3$ is just the volume of a sphere of radius d.) Expressing d in units of parsecs and writing it in terms of the apparent magnitude m (Eq. 24.1), we have

$$\begin{aligned}\overline{N}_M(M, S, \Omega, m) &= \frac{\Omega}{3}n_M(M, S)\,10^{3(m-M+5)/5}\\ &= \frac{\Omega}{3}n_M(M, S)\,e^{\ln 10^{3(m-M+5)/5}}\\ &= \frac{\Omega}{3}n_M(M, S)\,e^{[3(m-M+5)/5]\,\ln 10}.\end{aligned}$$